

# Writing a Simple Operating System — from Scratch

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# Chapter 1

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## Introduction

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We've all used an operating system (OS) before (e.g. Windows XP, Linux, etc.), and perhaps we have even written some programs to run on one; but what is an OS actually there for? how much of what I see when I use a computer is done by hardware and how much is done by software? and how does the computer actually work?

The late Prof. Doug Shepherd, a lively teacher of mine at Lancaster University, once reminded me amid my grumbling about some annoying programming problem that, back in the day, before he could even *begin* to attempt any research, he had to write his own operating system, from scratch. So it seems that, today, we take a lot for granted about how these wonderful machines actually work underneath all those layers of software that commonly come bundled with them and which are required for their day-to-day usefulness.

Here, concentrating on the widely used x86 architecture CPU, we will strip bare our computer of *all* software and follow in Doug's early footsteps, learning along the way about:

- How a computer boots
- How to write low-level programs in the barren landscape where no operating system yet exists
- How to configure the CPU so that we can begin to use its extended functionality
- How to bootstrap code written in a higher-level language, so that we can really start to make some progress towards our own operating system
- How to create some fundamental operating system services, such as device drivers, file systems, multi-tasking processing.

Note that, in terms of practical operating system functionality, this guide does not aim to be extensive, but instead aims to pool together snippets of information from many sources into a self-contained and coherent document, that will give you a hands-on experience of low-level programming, how operating systems are written, and the kind of problems they must solve. The approach taken by this guide is unique in that the particular languages and tools (e.g. assembly, C, Make, etc.) are not the focus but instead are treated as a means to an end: we will learn what we need to about these things to help us achieve our main goal.

This work is not intended as a replacement but rather as a stepping stone to excellent work such as the Minix project [?] and to operating system development in general.

## Chapter 2

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# Computer Architecture and the Boot Process

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## 2.1 The Boot Process

Now, we begin our journey.

When we reboot our computer, it must start up again, initially without any notion of an operating system. Somehow, it must load the operating system --- whatever variant that may be --- from some permanent storage device that is currently attached to the computer (e.g. a floppy disk, a hard disk, a USB dongle, etc.).

As we will shortly discover, the pre-OS environment of your computer offers little in the way of rich services: at this stage even a simple file system would be a luxury (e.g. read and write logical files to a disk), but we have none of that. Luckily, what we do have is the Basic Input/Output Software (BIOS), a collection of software routines that are initially loaded from a chip into memory and initialised when the computer is switched on. BIOS provides auto-detection and basic control of your computer's essential devices, such as the screen, keyboard, and hard disks.

After BIOS completes some low-level tests of the hardware, particularly whether or not the installed memory is working correctly, it must boot the operating system stored on one of your devices. Here, we are reminded, though, that BIOS cannot simply load a file that represents your operating system from a disk, since BIOS has no notion of a file-system. BIOS must read specific sectors of data (usually 512 bytes in size) from specific physical locations of the disk devices, such as Cylinder 2, Head 3, Sector 5 (details of disk addressing are described later, in Section XXX).

So, the easiest place for BIOS to find our OS is in the first sector of one of the disks (i.e. Cylinder 0, Head 0, Sector 0), known as the *boot sector*. Since some of our disks may not contain an operating systems (they may simply be connected for additional storage), then it is important that BIOS can determine whether the boot sector of a particular disk is boot code that is intended for execution or simply data. Note that the CPU does not differentiate between code and data: both can be interpreted as CPU instructions, where code is simply instructions that have been crafted by a programmer into some useful algorithm.

Again, an unsophisticated means is adopted here by BIOS, whereby the last two bytes of an intended boot sector must be set to the magic number `0xaa55`. So, BIOS loops through each storage device (e.g. floppy drive, hard disk, CD drive, etc.), reads the boot sector into memory, and instructs the CPU to begin executing the first boot sector it finds that ends with the magic number.

This is where we seize control of the computer.

## 2.2 BIOS, Boot Blocks, and the Magic Number

If we use a binary editor, such as TextPad [?] or GHex [?], that will let us write raw byte values to a file --- rather than a standard text editor that will convert characters such as 'A' into ASCII values --- then we can craft ourselves a simple yet valid boot sector.

```
e9 fd ff 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
*
00 00 00 00 00 00 00 00 00 00 00 00 00 00 55 aa
```

Figure 2.1: A machine code boot sector, with each byte displayed in hexadecimal.

Note that, in Figure 2.1, the three important features are:

- The initial three bytes, in hexadecimal as `0xe9`, `0xfd` and `0xff`, are actually machine code instructions, as defined by the CPU manufacturer, to perform an endless jump.
- The last two bytes, `0x55` and `0xaa`, make up the magic number, which tells BIOS that this is indeed a boot block and not just data that happens to be on a drive's boot sector.
- The file is padded with zeros ('\*' indicates zeros omitted for brevity), basically to position the magic BIOS number at the end of the 512 byte disk sector.

An important note on endianness. You might be wondering why the magic BIOS number was earlier described as the 16-bit value `0xaa55` but in our boot sector was written as the consecutive bytes `0x55` and `0xaa`. This is because the x86 architecture handles multi-byte values in *little-endian* format, whereby less significant bytes proceed more significant bytes, which is contrary to our familiar numbering system --- though if our system ever switched and I had £0000005 in my bank account, I would be able to retire now, and perhaps donate a couple of quid to the needy Ex-millionaires Foundation.

Compilers and assemblers can hide many issues of endianness from us by allowing us to define the types of data, such that, say, a 16-bit value is serialised automatically into machine code with its bytes in the correct order. However, it is sometimes useful,

especially when looking for bugs, to know exactly where an individual byte will be stored on a storage device or in memory, so endianness is very important.

This is possibly the smallest program your computer could run, but it is a valid program nonetheless, and we can test this in two ways, the second of which is much safer and better suited to our kind of experiments:

- Using whatever means your current operating system will allow, write this boot block to the first sector of a non-essential storage device (e.g. floppy disk or flash drive), then reboot the computer.
- Use virtual machine software, such as VMWare or VirtualBox, and set the boot block code as a disk image of a virtual machine, then start-up the virtual machine.

You can be sure this code has been loaded and executed if your computer simply hangs after booting, without a message such as “No operating system found”. This is the infinite loop at work, that we put at the start of the code. Without this loop the CPU would tear off, executing every subsequent instruction in memory, most of which will be random, uninitialised bytes, until it throws itself into some invalid state and either reboots or, by chance, stumbles upon and runs a BIOS routine that formats your main disk.

Remember, it is us that program the computer, and the computer follows our instructions blindly, fetching and executing them, until it is switched off; so we need to make sure that it executes our crafted code rather than random bytes of data held somewhere in memory. At this low level, we have a lot of power and responsibility over our computer, so we need to learn how to control it.

## 2.3 CPU Emulation

There is a *third*, more convenient option for testing these low-level programs without continuously having to reboot a machine or risk scrubbing your important data off a disk, and that is to use a CPU emulator such as Bochs or QEmu. Unlike machine virtualisation (e.g. VMware, VirtualBox), which tries to optimise for performance and therefore usage of the hosted operating system by running guest instructions directly on the CPU, emulation involves a program that behaves like a specific CPU architecture, using variables to represent CPU registers and high-level control structures to simulate lower level jumps and so on, so is much slower but often better suited for development and debugging such systems.

Note that, in order to do anything useful with an emulator, you need to give it some code to run in the form of a disk image file. An image file simply is the raw data (i.e. machine code and data) that would otherwise have been written to medium of a hard disk, a floppy disk, a CDROM, USB stick, etc. Indeed, some emulators will successfully boot and run a real operating system from an image file downloaded or extracted from an installation CDROM --- though virtualisation is better suited to this kind of use.

The emulators translate low-level display device instructions into pixel rendering on a desktop window, so you can see exactly what would be rendered on a real monitor.

In general, and for the exercises in this document, it follows that any machine code that runs correctly under an emulator will run correctly on the real architecture --- though obviously must faster.

### 2.3.1 Bochs: A x86 CPU Emulator

Bochs requires that we set up a simple configuration file, `bochsrc`, in the local directory, that describes details of how real devices (e.g. the screen and keyboard) are to be emulated and, importantly, which floppy disk image is to be booted when the emulated computer starts.

Figure 2.2 shows a sample Bochs configuration file that we can use to test the boot sector written in Section XXX and saved as the file `boot_sect.bin`

```
# Tell bochs to use our boot sector code as though it were
# a floppy disk inserted into a computer at boot time.
floppya: 1_44=boot_sect.bin, status=inserted
boot: a
```

Figure 2.2: A simple Bochs configuration file.

To test our boot sector in Bochs, simply type:

```
$bochs
```

As a simple experiment, try changing the BIOS magic number in our boot sector to something invalid then re-running Bochs.

Since Bochs' emulation of a CPU is close to the real thing, after you've tested code in Bochs, you should be able to boot it on a real machine, on which it will run much faster.

### 2.3.2 QEmu

QEmu is similar to Bochs, though is much more efficient and capable also of emulating architectures other than x86. Though QEmu is less well documented than Bochs, a need for no configuration file means it is easier to get running, as follows:

```
$qemu <your-os-boot-disk-image-file>
```

## 2.4 The Usefulness of Hexadecimal Notation

We've already seen some examples of *hexadecimal*, so it is important to understand why hexadecimal is often used in lower-level programming.

First it may be helpful to consider why counting in ten seems so natural to us, because when we see hexadecimal for the first time we always ask ourselves: why not simply count to ten? Not being an expert on the matter, I will make the assumption that counting to ten has something to do with most people having a total of ten fingers on their hands, which led to the ideas of numbers being represented as 10 distinct symbols: 0, 1, 2, ..., 8, 9

Decimal has a base of ten (i.e. has ten distinct digit symbols), but hexadecimal has a base of 16, so we have to invent some new number symbols; and the lazy way is just to use a few letters, giving us: `0,1,2,...,8,9,a,b,c,d,e,f`, where the single digit `d`, for example, represents a count of 13.

To distinguish among hexadecimal and other number systems, we often use the prefix `0x`, or sometimes the suffix `h`, which is especially important for hexadecimal digits that happen not to contain any of the letter digits, for example: `0x50` does not equal (decimal) `50` --- `0x50` is actually `80` in decimal.

The thing is, that a computer represent a number as a sequence of *bits* (binary digits), since fundamentally its circuitry can distinguish between only two electrical states: `0` and `1` --- it's like the computer has a total of only two fingers. So, to represent a number larger than `1`, the computer can bunch together a series of bits, just like we may count higher than `9` by having two or more digits (e.g. `456`, `23`, etc.).

Names have been adopted for bit series of certain lengths to make it easier to talk about and agree upon the size of numbers we are dealing with. The instructions of most computers deal with a minimum of 8 bit values, which are named *bytes*. Other groupings are *short*, *int*, and *long*, which usually represent 16-bit, 32-bit, and 64-bit values, respectively. We also see the term *word*, that is used to describe the size of the maximum processing unit of the current mode of the CPU: so in 16-bit real mode, a *word* refers to a 16-bit value; in 32-bit protected mode, a *word* refers to a 32-bit value; and so on.

So, returning to the benefit of hexadecimal: strings of bits are rather long-winded to write out but are much easier to convert to and from the more shorthand hexadecimal notation than to and from our natural decimal system, essentially because we can break the conversion down into smaller, 4-bit segments of the binary number, rather than try to add up all of the component bits into a grand total, which gets much harder for larger bit strings (e.g. 16, 32, 64, etc.). This difficulty with decimal conversion is shown clearly by the example given in Figure 2.3.

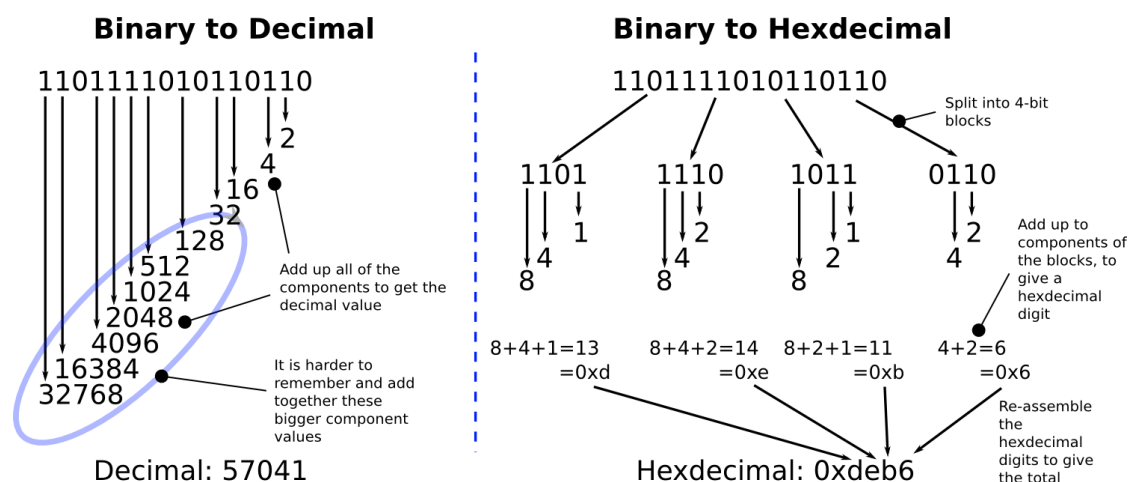


Figure 2.3: Conversion of 1101111010110110 to decimal and hexadecimal

## Chapter 3

---

# Boot Sector Programming (in 16-bit Real Mode)

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Even with the example code provided, you will no doubt have found it frustrating writing machine code in a binary editor. You'd have to remember, or continuously reference, which of many possible machine codes cause the CPU to do certain functions. Luckily, you are not alone, and so *assemblers* have been written that translate more human friendly instructions into machine code for a particular CPU.

In this chapter we will explore some increasingly sophisticated boot sector programs to familiarise ourselves with assembly and the barren, pre-OS environment in which our programs will run.

### 3.1 Boot Sector Re-visited

Now, we will re-create the binary-edited boot sector from Section XXX instead using assembly language, so that we can really appreciate the value even of a very low-level language.

We can assemble this into actual machine code (a sequence of bytes that our CPU can interpret as instructions) as follows:

```
$nasm boot_sect.asm -f bin -o boot_sect.bin
```

Where `boot_sect.asm` is the file into which we saved the source code in Figure 3.1 and `boot_sect.bin` is the assembled machine code that we can install as a boot sector on a disk.

Note that we used the `-f bin` option to instruct `nasm` to produce *raw* machine code, rather than a code package that has additional meta information for linking in other routines that we would expect to use when programming in a more typical operating system environment. We need none of that cruft. Apart from the low-level BIOS routines, we are the only software running on this computer now. We are the operating system now, albeit at this stage with nothing more to offer than an endless loop --- but we will soon build up from this.

```

;
; A simple boot sector program that loops forever.
;

loop:                ; Define a label, "loop", that will allow
                    ; us to jump back to it, forever.

    jmp loop         ; Use a simple CPU instruction that jumps
                    ; to a new memory address to continue execution.
                    ; In our case, jump to the address of the current
                    ; instruction.

times 510-($-$$) db 0 ; When compiled, our program must fit into 512 bytes,
                    ; with the last two bytes being the magic number,
                    ; so here, tell our assembly compiler to pad out our
                    ; program with enough zero bytes (db 0) to bring us to the
                    ; 510th byte.

dw 0xaa55            ; Last two bytes (one word) form the magic number,
                    ; so BIOS knows we are a boot sector.

```

Figure 3.1: A simple boot sector written in assembly language.

Rather than saving this to the boot sector of a floppy disk and rebooting our machine, we can conveniently test this program by running Bochs:

```
$bochs
```

Or, depending on our preference and on availability of an emulator, we could use QEmu, as follows:

```
$qemu boot_sect.bin
```

Alternatively, you could load the image file into virtualisation software or write it onto some bootable medium and boot it from a real computer. Note that, when you write an image file to some bootable medium, that does not mean you add the file to the medium's file system: you must use an appropriate tool to write directly to the medium in a low-level sense (e.g. directly to the sectors of a disk).

If we'd like to see more easily exactly what bytes the assembler created, we can run the following command, which displays the binary contents of the file in an easy-to-read hexadecimal format:

```
$od -t x1 -A n boot_sect.bin
```

The output of this command should look familiar.

Congratulations, you just wrote a boot sector in assembly language. As we will see, all operating systems must start this way and then pull themselves up into higher level abstractions (e.g. higher level languages, such as C/C++)

## 3.2 16-bit Real Mode

CPU manufacturers must go to great lengths to keep their CPUs (i.e. their specific instruction set) compatible with earlier CPUs, so that older software, and in particular older operating systems, can still run on the most modern CPUs.

The solution implemented by Intel and compatible CPUs is to *emulate* the oldest CPU in the family: the Intel *8086*, which had support for 16-bit instructions and no notion of *memory protection*: memory protection is crucial for the stability of modern operating systems, since it allows an operating system to restrict a user's process from accessing, say, kernel memory, which, whether done accidentally or on purpose, could allow such a process to circumvent security mechanisms or even bring down the whole system.

So, for backward compatibility, it is important that CPUs boot initially in *16-bit real mode*, requiring modern operating systems explicitly to switch up into the more advanced 32-bit (or 64-bit) protected mode, but allowing older operating systems to carry on, blissfully unaware that they are running on a modern CPU. Later on, we will look at this important step from 16-bit real mode into 32-bit protected mode in detail.

Generally, when we say that a CPU is 16-bit, we mean that its instructions can work with a maximum of 16-bits at once, for example: a 16-bit CPU will have a particular instruction that can add two 16-bit numbers together in one CPU cycle; if it was necessary for a process to add together two 32-bit numbers, then it would take more cycles, that make use of 16-bit addition.

First we will explore this 16-bit real mode environment, since all operating systems must begin here, then later we will see how to switch into 32-bit protected mode and the main benefits of doing so.

## 3.3 Erm, Hello?

Now we are going to write a *seemingly* simple boot sector program that prints a short message on the screen. To do this we will have to learn some fundamentals of how the CPU works and how we can use BIOS to help us to manipulate the screen device.

Firstly, let's think about what we are trying to do here. We'd like to print a character on the screen but we do not know exactly how to communicate with the screen device, since there may be many different kinds of screen devices and they may have different interfaces. This is why we need to use BIOS, since BIOS has already done some auto detection of the hardware and, evidently by the fact that BIOS earlier printed information on the screen about self-testing and so on, so can offer us a hand.

So, next, we'd like to ask BIOS to print a character for us, but how do we ask BIOS to do that? There are no Java libraries for printing to the screen --- they are a dream away. We can be sure, however, that somewhere in the memory of the computer there will be some BIOS machine code that knows how to write to the screen. The truth is that we could possibly find the BIOS code in memory and execute it somehow, but this is more trouble than it is worth and will be prone to errors when there are differences between BIOS routine internals on different machines.

Here we can make use of a fundamental mechanism of the computer: *interrupts*.

### 3.3.1 Interrupts

Interrupts are a mechanism that allow the CPU temporarily to halt what it is doing and run some other, higher-priority instructions before returning to the original task. An interrupt could be raised either by a software instruction (e.g. `int 0x10`) or by some hardware device that requires high-priority action (e.g. to read some incoming data from a network device).

Each interrupt is represented by a unique number that is an index to the interrupt vector, a table initially set up by BIOS at the start of memory (i.e. at physical address `0x0`) that contains address pointers to *interrupt service routines* (ISRs). An ISR is simply a sequence of machine instructions, much like our boot sector code, that deals with a specific interrupt (e.g. perhaps to read new data from a disk drive or from a network card).

So, in a nutshell, BIOS adds some of its own ISRs to the interrupt vector that specialise in certain aspects of the computer, for example: interrupt `0x10` causes the screen-related ISR to be invoked; and interrupt `0x13`, the disk-related I/O ISR.

However, it would be wasteful to allocate an interrupt per BIOS routine, so BIOS multiplexes the ISRs by what we could imagine as a big `switch` statement, based usually on the value set in one of the CPUs general purpose registers, `ax`, prior to raising the interrupt.

### 3.3.2 CPU Registers

Just as we use variables in a higher level languages, it is useful if we can store data temporarily during a particular routine. All x86 CPUs have four general purpose *registers*, `ax`, `bx`, `cx`, and `dx`, for exactly that purpose. Also, these registers, which can each hold a *word* (two bytes, 16 bits) of data, can be read and written by the CPU with negligible delay as compared with accessing main memory. In assembly programs, one of the most common operations is moving (or more accurately, *copying*) data between these registers:

```
mov ax, 1234      ; store the decimal number 1234 in ax
mov cx, 0x234     ; store the hex number 0x234 in cx
mov dx, 't'       ; store the ASCII code for letter 't' in dx
mov bx, ax        ; copy the value of ax into bx, so now bx == 1234
```

Notice that the destination is the first and not second argument of the `mov` operation, but this convention varies with different assemblers.

Sometimes it is more convenient to work with single bytes, so these registers let us set their high and low bytes independently:

```
mov ax, 0          ; ax -> 0x0000, or in binary 0000000000000000
mov ah, 0x56       ; ax -> 0x5600
mov al, 0x23       ; ax -> 0x5623
mov ah, 0x16       ; ax -> 0x1623
```

[?]

### 3.3.3 Putting it all Together

So, recall that we'd like BIOS to print a character on the screen for us, and that we can invoke a specific BIOS routine by setting `ax` to some BIOS-defined value and then

triggering a specific interrupt. The specific routine we want is the BIOS scrolling teletype routine, which will print a single character on the screen and advance the cursor, ready for the next character. There is a whole list of BIOS routines published that show you which interrupt to use and how to set the registers prior to the interrupt. Here, we need interrupt `0x10` and to set `ah` to `0x0e` (to indicate teletype mode) and `al` to the ASCII code of the character we wish to print.

```
;
; A simple boot sector that prints a message to the screen using a BIOS routine.
;

mov ah, 0x0e ; int 10/ah = 0eh -> scrolling teletype BIOS routine

mov al, 'H'
int 0x10
mov al, 'e'
int 0x10
mov al, 'l'
int 0x10
mov al, 'l'
int 0x10
mov al, 'o'
int 0x10

jmp $ ; Jump to the current address (i.e. forever).

;
; Padding and magic BIOS number.
;

times 510-($-$$) db 0 ; Pad the boot sector out with zeros

dw 0xaa55 ; Last two bytes form the magic number,
; so BIOS knows we are a boot sector.
```

Figure 3.2:

Figure 3.2 shows the whole boot sector program. Notice how, in this case, we only needed to set `ah` once, then just changed `al` for different characters.

```
b4 0e b0 48 cd 10 b0 65 cd 10 b0 6c cd 10 b0 6c
cd 10 b0 6f cd 10 e9 fd ff 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
*
00 00 00 00 00 00 00 00 00 00 00 00 00 55 aa
```

Figure 3.3:

Just for completeness, Figure 3.3 shows the raw machine code of this boot sector. These are the actual bytes that are telling the CPU exactly what to do. If you are surprised by the amount of effort and understanding that is involved in writing such a barely --- if at all --- useful program, then remember that these instructions map very closely to the CPU's circuitry, so necessarily they are very simple, but also very fast. You are getting to know your computer now, as it really *is*.

## 3.4 Hello, World!

Now we are going to attempt a slightly more advanced version of the 'hello' program, that introduces a few more CPU fundamentals and an understanding of the landscape of memory into which our boot sector gets plonked by BIOS.

### 3.4.1 Memory, Addresses, and Labels

We said earlier how the CPU fetches and executes instructions from memory, and how it was BIOS that loaded our 512-byte boot sector into memory and then, having finished its initialisations, told the CPU to jump to the start of our code, whereupon it began executing our first instruction, then the next, then the next, etc.

So our boot sector code is somewhere in memory; but where? We can imagine the main memory as long sequence of bytes that can individually be accessed by an address (i.e. an index), so if we want to find out what is in the 54th byte of memory, then 54 is our address, which is often more convenient to express in hexadecimal: `0x36`.

So the start of our boot-sector code, the very first machine code byte, is at some address in memory, and it was BIOS that put us there. We might assume, unless we knew otherwise, that BIOS loaded our code at the start of memory, at address `0x0`. It's not so straightforward, though, because we know that BIOS has already been doing initialisation work on the computer long before it loaded our code, and will actually continue to service hardware interrupts for the clock, disk drives, and so on. So these BIOS routines (e.g. ISRs, services for screen printing, etc.) themselves must be stored somewhere in memory and must be preserved (i.e. not overwritten) whilst they are still of use. Also, we noted earlier that the interrupt vector is located at the start of memory, and were BIOS to load us there, our code would stomp over the table, and upon the next interrupt occurring, the computer will likely crash and reboot: the mapping between interrupt number and ISR would effectively have been severed.

As it turns out, BIOS likes always to load the boot sector to the address `0x7c00`, where it is sure will not be occupied by important routines. Figure 3.4 gives an example of the typical low memory layout of the computer when our boot sector has just been loaded [?]. So whilst we may instruct the CPU to write data to any address in memory, it may cause bad things to happen, since some memory is being used by other routines, such as the timer interrupt and disk devices.

### 3.4.2 'X' Marks the Spot

Now we are going to play a game called "find the byte", which will demonstrate memory referencing, the use of labels in assembly code, and the importance of knowing where BIOS loaded us to. We are going to write an assembly program that reserves a byte of

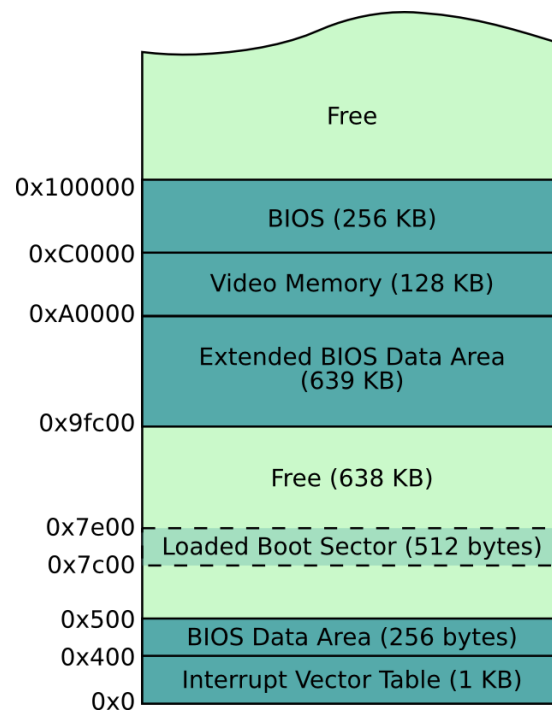


Figure 3.4: Typical lower memory layout after boot.

data for a character, then we will try to print out that character on the screen. To do this we need to figure out its absolute memory address, so we can load it into `al` and get BIOS to print it, as in the last exercise.

```
;
; A simple boot sector program that demonstrates addressing.
;
mov ah, 0x0e          ; int 10/ah = 0eh -> scrolling teletype BIOS routine

; First attempt
mov al, the_secret
int 0x10              ; Does this print an X?

; Second attempt
mov al, [the_secret]
int 0x10              ; Does this print an X?

; Third attempt
mov bx, the_secret
add bx, 0x7c00
mov al, [bx]
int 0x10              ; Does this print an X?

; Fourth attempt
```

```

mov al, [0x7c1e]
int 0x10          ; Does this print an X?

jmp $             ; Jump forever.

the_secret:
db "X"

; Padding and magic BIOS number.

times 510-($-$$) db 0
dw 0xaa55

```

Firstly, when we declare some data in our program, we prefix it with a label (`the_secret`). We can put labels anywhere in our programs, with their only purpose being to give us a convenient offset from the start of the code to a particular instruction or data.

```

b4 0e b0 1e cd 10 a0 1e 00 cd 10 bb 1e 00 81 c3
00 7c 8a 07 cd 10 a0 1e 7c cd 10 e9 fd ff 58 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
*
00 00 00 00 00 00 00 00 00 00 00 00 00 55 aa

```

Figure 3.5:

If we look at the assembled machine code in Figure 3.5, we can see that our 'X', which has an hexadecimal ASCII code `0x58`, is at an offset of 30 (`0x1e`) bytes from the start of the code, immediately before we padded the boot sector with zeros.

If we run the program we see that only the second two attempts succeed in printing an 'X'.

The problem with the first attempt is that it tries to load the immediate offset into `al` as the character to print, but actually we wanted to print the character *at* the offset rather than the offset itself, as attempted next, whereby the square brackets instruct the CPU to do this very thing - store the *contents* of an address.

So why does the second attempt fail? The problem is, that the CPU treats the offset as though it was from the start of memory, rather than the start address of our loaded code, which would land it around about in the interrupt vector. In the third attempt, we add the offset `the_secret` to the address that we believe BIOS to have loaded our code, `0x7c00`, using the CPU `add` instruction. We can think of `add` as the higher level language statement `bx = bx + 0x7c00`. We have now calculated the correct memory address of our 'X' and can store the contents of that address in `al`, ready for the BIOS print function, with the instruction `mov al, [bx]`.

In the fourth attempt we try to be a bit clever, by pre-calculating the address of the 'X' after the boot sector is loaded into memory by BIOS. We arrive at the address `0x7c1e` based on our earlier examination of the binary code (See Figure 3.5) which revealed that 'X' was `0x1e` (30) bytes from the start of our boot sector. This last example reminds

us why labels are useful, since without labels we would have to count offsets from the compiled code, and then update these when changes in code cause these offsets to change.

So now we have seen how BIOS does indeed load our boot sector to the address `0x7c00`, and we have also seen how addressing and assembly code labels are related.

It is inconvenient to always have to account for this label--memory offset in your code, so many assemblers will correct label references during assembleage if you include the following instruction at the top of your code, telling it exactly where you expect the code to loaded in memory:

```
[org 0x7c00]
```

### Question 1

What do you expect will be printed now, when this `org` directive is added to this boot-sector program? For good marks, explain why this is so.

### 3.4.3 Defining Strings

Supposing you wanted to print a pre-defined message (e.g. “Booting OS”) to the screen at some point; how would you define such a string in your assembly program? We have to remind ourselves that our computer knows nothing about strings, and that a string is merely a sequence of data units (e.g. bytes, words, etc.) held somewhere in memory.

In the assembler we can define a string as follows:

```
my_string:
    db 'Booting OS'
```

We’ve actually already seen `db`, which translates to “declare byte(s) of data”, which tells the assembler to write the subsequent bytes directly to the binary output file (i.e. do not interpret them as processor instructions). Since we surrounded our data with quotes, the assembler knows to convert each character to its ASCII byte code. Note that, we often use a label (e.g. `my_string`) to mark the start of our data, otherwise we would have no easy way of referencing it within our code.

One thing we have overlooked in this example is that knowing how *long* a string is equally important as to knowing where it is. Since it is us that has to write all the code that handles strings, it is important to have a consistent strategy for knowing how long a string is. There are a few possibilities, but the convention is to declare strings as *null-terminating*, which means we always declare the last byte of the string as `0`, as follows:

```
my_string:
    db 'Booting OS',0
```

When later iterating through a string, perhaps to print each of its characters in turn, we can easily determine when we have reached the end.

### 3.4.4 Using the Stack

When on the topic of low-level computing, we often hear people talking about the *stack* like it is some special thing. The stack is really just a simple solution to the following inconvenience: the CPU has a limited number of registers for the temporary storage of our routine's local variables, but we often need more temporary storage than will fit into these registers; now, we can obviously make use of main memory, but specifying specific memory addresses when reading and writing is inconvenient, especially since we do not care exactly where the data is to be stored, only that we can retrieve it easily enough. And, as we shall see later, the stack is also useful for argument passing to realise function calls.

So, the CPU offers two instructions `push` and `pop` that allow us, respectively, to store a value and retrieve a value from the top of the stack, and so without worrying exactly where they are stored. Note, however, that we cannot push and pop single bytes onto and off the stack: in 16-bit mode, the stack works only on 16-bit boundaries.

The stack is implemented by two special CPU registers, `bp` and `sp`, which maintain the addresses of the stack base (i.e. the stack bottom) and the stack top respectively. Since the stack expands as we push data onto it, we usually set the stack's base far away from important regions of memory (e.g. such as BIOS code or our code) so there is no danger of overwriting if the stack grows too large. One confusing thing about the stack is that it actually grows *downwards* from the base pointer, so when we issue a `push`, the value actually gets stored below --- and not above --- the address of `bp`, and `sp` is decremented by the value's size.

The following boot sector program in Figure 3.6 demonstrates use of the stack.

#### Question 2

What will be printed in what order by the code in Figure 3.6? And at what absolute memory address will the ASCII character 'C' be stored? You may find it useful to modify the code to confirm your expectation, but be sure to explain *why* it is this address.

### 3.4.5 Control Structures

We'd never be comfortable using a programming language if we didn't know how to write some basic control structures, such as `if..then..elseif..else`, `for`, and `while`. These structures allow alternative branches of execution and form the basis of any useful routine.

After compilation, these high-level control structures reduce to simple jump statements. Actually, we've already seen the simplest example of loops:

```
some_label:
    jmp some_label    ; jump to address of label
```

Or alternatively, with identical effect:

```
    jmp $              ; jump to address of current instruction
```

So this instruction offers us an *unconditional* jump (i.e. it will *always* jump); but we often need to jump based on some condition (e.g. carry on looping *until we have looped ten times*, etc.).

```

;
; A simple boot sector program that demonstrates the stack.
;
mov ah, 0x0e      ; int 10/ah = 0eh -> scrolling teletype BIOS routine

mov bp, 0x8000    ; Set the base of the stack a little above where BIOS
mov sp, bp        ; loads our boot sector - so it won't overwrite us.

push 'A'          ; Push some characters on the stack for later
push 'B'          ; retrieval. Note, these are pushed on as
push 'C'          ; 16-bit values, so the most significant byte
                  ; will be added by our assembler as 0x00.

pop bx            ; Note, we can only pop 16-bits, so pop to bx
mov al, bl        ; then copy bl (i.e. 8-bit char) to al
int 0x10          ; print(al)

pop bx            ; Pop the next value
mov al, bl        ;
int 0x10          ; print(al)

mov al, [0x7ffe]  ; To prove our stack grows downwards from bp,
                  ; fetch the char at 0x8000 - 0x2 (i.e. 16-bits)
int 0x10          ; print(al)

jmp $             ; Jump forever.

; Padding and magic BIOS number.

times 510-($-$$) db 0
dw 0xaa55

```

Figure 3.6: Manipulation of the stack, using `push` and `pop`

Conditional jumps are achieved in assembly language by first running a comparison instruction, then by issuing a specific conditional jump instruction.

```

cmp ax, 4         ; compare the value in ax to 4
je then_block     ; jump to then_block if they were equal
mov bx, 45        ; otherwise, execute this code
jmp the_end       ; important: jump over the 'then' block,
                  ; so we don't also execute that code.

then_block:
mov bx, 23
the_end:

```

In a language such as C or Java, this would look like this:

```

if(ax == 4) {
    bx = 23;
} else {
    bx = 45;
}

```

We can see from the assembly example that there is something going on behind the scenes that is relating the `cmp` instruction to the `je` instruction it proceeds. This is an example of where the CPU's special `flags` register is used to capture the outcome of the `cmp` instruction, so that a subsequent conditional jump instruction can determine whether or not to jump to the specified address.

The following jump instructions are available, based on an earlier `cmp x, y` instruction:

```
je target    ; jump if equal           (i.e. x == y)
jne target   ; jump if not equal       (i.e. x != y)
jl target    ; jump if less than       (i.e. x < y)
jle target   ; jump if less than or equal (i.e. x <= y)
jg target    ; jump if greater than    (i.e. x > y)
jge target   ; jump if greater than or equal (i.e. x >= y)
```

### Question 3

It's always useful to plan your conditional code in terms of a higher level language, then replace it with the assembly instructions. Have a go at converting this pseudo assembly code into full assembly code, using `cmp` and appropriate jump instructions. Test it with different values of `bx`. Fully comment your code, in your own words.

```
mov bx, 30

if (bx <= 4) {
    mov al, 'A'
} else if (bx < 40) {
    mov al, 'B'
} else {
    mov al, 'C'
}

mov ah, 0x0e    ; int=10/ah=0x0e -> BIOS tele-type output
int 0x10        ; print the character in al

jmp $

; Padding and magic number.
times 510-($-$$) db 0
dw 0xaa55
```

### 3.4.6 Calling Functions

In high-level languages, we break big problems down into functions, which essentially are general purpose routines (e.g. print a message, write to a file, etc.) that we use over and over again throughout our program, usually changing parameters that we pass to the function to change the outcome in some way. At the CPU level a function is nothing more than a jump to the address of a useful routine then a jump back again to the instruction immediately following the first jump.

We can kind of simulate a function call like this:

```
...
...
mov al, 'H'           ; Store 'H' in al so our function will print it.
```

```

    jmp my_print_function
return_to_here:    ; This label is our life-line so we can get back.
    ...
    ...

my_print_function:
    mov ah, 0x0e    ; int=10/ah=0x0e -> BIOS tele-type output
    int 0x10        ; print the character in al
    jmp return_to_here ; return from the function call.

```

Firstly, note how we used the register `al` as a parameter, by setting it up ready for the function to use. This is how parameter passing is made possible in higher level languages, where the *caller* and *callee* must have some agreement on where and how many parameters will be passed.

Sadly, the main flaw with this approach is that we need to say explicitly where to return to after our function has been called, and so it will not be possible to call this function from arbitrary points in our program --- it will always return the same address, in this case the label `return_to_here`.

Borrowing from the parameter passing idea, the caller code could store the correct return address (i.e. the address immediately after the call) in some well-known location, then the called code could jump back to that stored address. The CPU keeps track of the current instruction being executed in the special register `ip` (instruction pointer), which, sadly, we cannot access directly. However, the CPU provides a pair of instructions, `call` and `ret`, which do exactly what we want: `call` behaves like `jmp` but additionally, before actually jumping, pushes the return address on to the stack; `ret` then pops the return address off the stack and jumps to it, as follows:

```

    ...
    ...
    mov al, 'H'    ; Store 'H' in al so our function will print it.
    call my_print_function
    ...
    ...

my_print_function:
    mov ah, 0x0e    ; int=10/ah=0x0e -> BIOS tele-type output
    int 0x10        ; print the character in al
    ret

```

Our functions are almost self-contained now, but there is still an ugly problem that we will thank ourselves later for if we now take the trouble to consider it. When we call a function, such as a print function, within our assembly program, internally that function may alter the values of several registers to perform its job (indeed, with registers being a scarce resource, it will almost certainly do this), so when our program returns from the function call it may not be safe to assume, say, the value we stored in `dx` will still be there.

It is often sensible (and polite), therefore, for a function immediately to push any registers it plans to alter onto the stack and then pop them off again (i.e. restore the registers' original values) immediately before it returns. Since a function may use many of the general purpose registers, the CPU implements two convenient instructions, `pusha` and `popa`, that conveniently push and pop *all* registers to and from the stack respectively, for example:

```

    ...

```

```

...

some_function:
    pusha                ; Push all register values to the stack
    mov bx, 10
    add bx, 20
    mov ah, 0x0e         ; int=10/ah=0x0e -> BIOS tele-type output
    int 0x10             ; print the character in al
    popa                ; Restore original register values
    ret

```

### 3.4.7 Include Files

After slaving away even on the seemingly simplest of assembly routines, you will likely want to reuse your code in multiple programs. nasm allows you to include external files literally as follows:

```

%include "my_print_function.asm" ; this will simply get replaced by
                                ; the contents of the file

...
mov al, 'H'                    ; Store 'H' in al so our function will print it.
call my_print_function

```

### 3.4.8 Putting it all Together

We now have enough knowledge about the CPU and assembly to write a more sophisticated “Hello, World” boot sector program.

#### Question 4

Put together all of the ideas in this section to make a self-contained function for printing null-terminated strings, that can be used as follows:

```

;
; A boot sector that prints a string using our function.
;
[org 0x7c00] ; Tell the assembler where this code will be loaded

    mov bx, HELLO_MSG ; Use BX as a parameter to our function, so
    call print_string ; we can specify the address of a string.

    mov bx, GOODBYE_MSG
    call print_string

    jmp $ ; Hang

%include "print_string.asm"

; Data
HELLO_MSG:
    db 'Hello, World!', 0 ; <-- The zero on the end tells our routine

```

```

                                ;      when to stop printing characters.
GOODBYE_MSG:
    db 'Goodbye!', 0

; Padding and magic number.
    times 510-($-$$) db 0
    dw 0xaa55

```

For good marks, make sure the function is careful when modifying registers and that you fully comment the code to demonstrate your understanding.

### 3.4.9 Summary

Still, it feels that we have not come very far. That's okay, and that's quite normal, given the primitive environment that we have been working in. If you have understood all up until here, then we are well on our way.

## 3.5 Nurse, Fetch me my Steth-o-scope

So far we have managed to get the computer to print out characters and strings that we have loaded into memory, but soon we will be trying to load some data from the disk, so it will be very helpful if we can display the hexadecimal values stored at arbitrary memory addresses, to confirm if we have indeed managed to load anything. Remember, we do not have the luxury of a nice development GUI, complete with a debugger that will let us carefully step through and inspect our code, and the best feedback the computer can give us when we make a mistake is visibly to do nothing at all, so we need to look after ourselves.

We have already written a routine to print out a string of characters, so we will now extend that idea into a hexadecimal printing routine --- a routine certainly to be cherished in this unforgiving, low-level world.

Let's think carefully about how we will do this, starting by considering how we'd like to use the routine. In a high-level language, we'd like something like this: `print_hex(0x1fb6)`, which would result in the string `'0x1fb6'` being printed on the screen. We have already seen, in Section XXX, how functions can be called in assembly and how we can use registers as parameters, so let's use the `dx` register as a parameter to hold the value we wish our `print_hex` function to print:

```

    mov dx, 0x1fb6    ; store the value to print in dx
    call print_hex    ; call the function

; prints the value of DX as hex.
print_hex:
    ...
    ...
    ret

```

Since we are printing a string to the screen, we might as well re-use our earlier printing function to do the actual printing part, then our main task is to look at how we can build that string from the value in our parameter, `dx`. We definitely don't want to confuse matters more than we need to when working in assembly, so let's consider the following trick to get us started with this function. If we define the complete hexadecimal string as a sort of template variable in our code, as we defined our earlier "Hello, World" messages, we can simply get the string printing function to print it, then the task of our `print_hex` routine is to alter the components of that template string to reflect the hexadecimal value as ASCII codes:

```

    mov dx, 0x1fb6    ; store the value to print in dx
    call print_hex    ; call the function

; prints the value of DX as hex.
print_hex:
    ; TODO: manipulate chars at HEX_OUT to reflect DX

    mov bx, HEX_OUT   ; print the string pointed to
    call print_string ; by BX
    ret

; global variables
HEX_OUT: db '0x0000',0

```

### 3.5.1 Question 5 (Advanced)

Complete the implementation of the `print_hex` function. You may find the CPU instructions `and` and `shr` to be useful, which you can find information about on the Internet. Make sure to fully explain your code with comments, in your own words.

## 3.6 Reading the Disk

We have now been introduced to BIOS, and have had a little play in the computer's low-level environment, but we have a little problem that poses to get in the way of our plan to write an operating system: BIOS loaded our boot code from the first sector of the disk, but that is *all* it loaded; what if our operating system code is larger --- and I'm guessing it will be --- than 512 bytes.

Operating systems usually don't fit into a single (512 byte) sector, so one of the first things they must do is bootstrap the rest of their code from the disk into memory and then begin executing that code. Luckily, as was hinted at earlier, BIOS provides routines that allow us to manipulate data on the drives.

### 3.6.1 Extended Memory Access Using Segments

When the CPU runs in its initial 16-bit real mode, the maximum size of the registers is 16 bits, which means that the highest address we can reference in an instruction is `0xffff`, which amounts by today's standards to a measly 64 KB (65536 bytes). Now, perhaps the likes of our intended simple operating system would not be affected by this limit,

but a day-to-day operating systems would never sit comfortably in such a tight box, so it is important that we understand the solution, of segmentation, to this problem.

To get around this limitation, the CPU designers added a few more special registers, `cs`, `ds`, `ss`, and `es`, called *segment* registers. We can imagine main memory as being divided into *segments* that are indexed by the segment registers, such that, when we specify a 16-bit address, the CPU automatically calculates the absolute address as the appropriate segment's start address offsetted by our specified address [?]. By *appropriate segment*, I mean that, unless explicitly told otherwise, the CPU will offset our address from the segment register appropriate for the context of our instruction, for example: the address used in the instruction `mov ax, [0x45ef]` would by default be offset from the *data segment*, indexed by `ds`; similarly, the *stack segment*, `ss`, is used to modify the actual location of the stack's base pointer, `bp`.

The most confusing thing about segment addressing is that adjacent segments overlap almost completely but for 16 bytes, so different segment and offset combinations can actually point to the same physical address; but enough of the talk: we won't truly grasp this concept until we've seen some examples.

To calculate the absolute address the CPU multiplies the value in the segment register by 16 and then adds your offset address; and because we are working with hexadecimal, when we multiple a number by 16, we simply shift it a digit to the left (e.g.  $0x42 * 16 = 0x420$ ). So if we set `ds` to `0x4d` and then issue the statement `mov ax, [0x20]`, the value stored in `ax` will actually be loaded from address `0x4d0` ( $16 * 0x4d + 0x20$ ).

Figure 3.7 shows how we can set `ds` to achieve a similar correction of label addressing as when we used the `[org 0x7c00]` directive in Section XXX. Because we do not use the `org` directive, the assembler does not offset our labels to the correct memory locations when the code is loaded by BIOS to the address `0x7c00`, so the first attempt to print an 'X' will fail. However, if we set the data segment register to `0x7c0`, the CPU will do this offset for us (i.e.  $0x7c0 * 16 + \text{the\_secret}$ ), and so the second attempt will correctly print the 'X'. In the third and fourth attempts we do the same, and get the same results, but instead explicitly state to the CPU which segment register to use when computing the physical address, using instead the general purpose segment register `es`.

Note that limitations of the CPU's circuitry (at least in 16-bit real mode) reveal themselves here, when seemingly correct instructions like `mov ds, 0x1234` are not actually possible: just because we can store a literal address directly into a general purpose register (e.g. `mov ax, 0x1234` or `mov cx, 0xdf`), it doesn't mean we can do the same with every type of register, such as segment registers; and so, as in Figure 3.7, we must take an additional step to transfer the value via a general purpose register.

So, segment-based addressing allows us to reach further into memory, up to a little over 1 MB ( $0xffff * 16 + 0xffff$ ). Later, we will see how more memory can be accessed, when we switch to 32-bit protected mode, but for now it suffices for us to understand 16-bit real mode segment-based addressing.

### 3.6.2 How Disk Drives Work

Mechanically, hard disk drives contain one or more stacked platters that spin under a read/write head, much like an old record player, only potentially, to increase capacity, with several records stacked one above the other, where a head moves in and out to get coverage of the whole of a particular spinning platter's surface; and since a particular platter may be readable and writable on both of its surfaces, one read/write head may

```

;
; A simple boot sector program that demonstrates segment offsetting
;
mov ah, 0x0e          ; int 10/ah = 0eh -> scrolling teletype BIOS routine

mov al, [the_secret]
int 0x10              ; Does this print an X?

mov bx, 0x7c0         ; Can't set ds directly, so set bx
mov ds, bx           ; then copy bx to ds.
mov al, [the_secret]
int 0x10              ; Does this print an X?

mov al, [es:the_secret] ; Tell the CPU to use the es (not ds) segment.
int 0x10              ; Does this print an X?

mov bx, 0x7c0
mov es, bx
mov al, [es:the_secret]
int 0x10              ; Does this print an X?

jmp $                ; Jump forever.

the_secret:
db "X"

; Padding and magic BIOS number.
times 510-($-$$) db 0
dw 0xaa55

```

Figure 3.7: Manipulating the data segment with the `ds` register.

float above and another below it. Figure 3.8 shows the inside of a typical hard disk drive, with the stack of platters and heads exposed. Note that the same idea applies to floppy disk drives, which, instead of several stacked hard platters, usually have a single, two-sided floppy disk medium.

The metallic coating of the platters give them the property that specific areas of their surface can be magnetised or demagnetised by the head, effectively allowing any state to be recorded permanently on them [?]. It is therefore important to be able to describe the exact place on the disk's surface where some state is to be read or written, and so Cylinder-Head-Sector (CHS) addressing is used, which effectively is a 3D coordinate system (see Figure 3.9):

- Cylinder: the cylinder describes the head's discrete distance from the outer edge of the platter and is so named since, when several platters are stacked up, you can visualise that all of the heads select a cylinder through all of the platters
- Head: the head describes which track (i.e. which specific platter surface within the cylinder) we are interested in.
- Sector: the circular track is divided into sectors, usually of capacity 512 bytes, which can be referenced with a sector index.



Figure 3.8: Inside of a hard disk drive

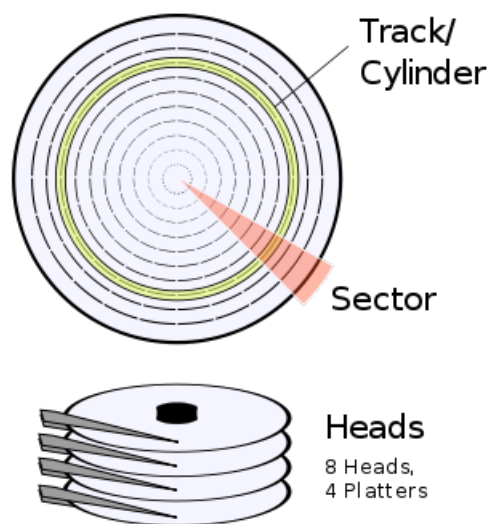


Figure 3.9: Cylinder, Head, Sector structure of a hard disk.

### 3.6.3 Using BIOS to Read the Disk

As we will see a little later on, specific devices require specific routines to be written to use them, so, for example, a floppy disk device requires us to explicitly turn on and off the motor that spins the disk under the read-and-write head before we can use it, whereas most hard disk devices have more functionality automated on local chips [?], but again the bus technologies with which such devices connect to the CPU (e.g. ATA/IDE, SATA, SCSI, USB, etc.) affect how we access them. Thankfully, BIOS can offer a few disk routines that abstract all of these differences for common disk devices.

The specific BIOS routine we are interested in here is accessed by raising interrupt `0x13` after setting the register `al` to `0x02`. This BIOS routine expects us to set up a few other registers with details of which disk device to use, which blocks we wish to read from the disk, and where to store the blocks in memory. The most difficult part of using this routine is that we must specify the first block to be read using a CHS addressing scheme; otherwise, it is just a case of filling in the expected registers, as detailed in the next code snippet.

```
mov ah, 0x02 ; BIOS read sector function

mov dl, 0    ; Read drive 0 (i.e. first floppy drive)
mov ch, 3    ; Select cylinder 3
mov dh, 1    ; Select the track on 2nd side of floppy
              ; disk, since this count has a base of 0
mov cl, 4    ; Select the 4th sector on the track - not
              ; the 5th, since this has a base of 1.
mov al, 5    ; Read 5 sectors from the start point

; Lastly, set the address that we'd like BIOS to read the
; sectors to, which BIOS expects to find in ES:BX
; (i.e. segment ES with offset BX).
mov bx, 0xa000 ; Indirectly set ES to 0xa000
mov es, bx
mov bx, 0x1234 ; Set BX to 0x1234
; In our case, data will be read to 0xa000:0x1234, which the
; CPU will translate to physical address 0xa1234

int 0x13      ; Now issue the BIOS interrupt to do the actual read.
```

Note that, for one reason or another (e.g. we indexed a sector beyond the limit of the disk, an attempt was made to read a faulty sector, the floppy disk was not inserted into the drive, etc.), BIOS may fail to read the disk for us, so it is important to know how to detect this; otherwise, we may *think* we have read some data but in fact the target address simply contains the same random bytes it did before we issued the read command. Fortunately for us, BIOS updates some registers to let us know what happened: the carry flag (CF) of the special `flags` register is set to signal a general fault, and `al` is set to the number of sectors *actually* read, as opposed to the number requested. After issuing the interrupt for the BIOS disk read, we can perform a simple check as follows:

```
...
...
int 0x13      ; Issue the BIOS interrupt to do the actual read.

jc disk_error ; jc is another jumping instruction, that jumps
              ; only if the carry flag was set.

; This jumps if what BIOS reported as the number of sectors
```

```

; actually read in AL is not equal to the number we expected.
cmp al, <no. sectors expected>
jne disk_error

disk_error :
    mov bx, DISK_ERROR_MSG
    call print_string
    jmp $

; Global variables
DISK_ERROR_MSG: db "Disk read error!", 0

```

### 3.6.4 Putting it all Together

As explained earlier, being able to read more data from the disk will be essential for bootstrapping our operating system, so here we will put all of the ideas from this section into a helpful routine that will simply read the first  $n$  sectors following the boot sector from a specified disk device.

```

; load DH sectors to ES:BX from drive DL
disk_load:
    push dx                ; Store DX on stack so later we can recall
                           ; how many sectors were request to be read,
                           ; even if it is altered in the meantime

    mov ah, 0x02           ; BIOS read sector function
    mov al, dh             ; Read DH sectors
    mov ch, 0x00           ; Select cylinder 0
    mov dh, 0x00           ; Select head 0
    mov cl, 0x02           ; Start reading from second sector (i.e.
                           ; after the boot sector)

    int 0x13              ; BIOS interrupt

    jc disk_error          ; Jump if error (i.e. carry flag set)

    pop dx                ; Restore DX from the stack
    cmp dh, al            ; if AL (sectors read) != DH (sectors expected)
    jne disk_error        ; display error message
    ret

disk_error :

    mov bx, DISK_ERROR_MSG
    call print_string
    jmp $

; Variables
DISK_ERROR_MSG db "Disk read error!", 0

```

And to test this routine, we can write a boot sector program as follows:

```

; Read some sectors from the boot disk using our disk_read function
[org 0x7c00]

    mov [BOOT_DRIVE], dl ; BIOS stores our boot drive in DL, so it's
                        ; best to remember this for later.

    mov bp, 0x8000      ; Here we set our stack safely out of the
    mov sp, bp          ; way, at 0x8000

    mov bx, 0x9000      ; Load 5 sectors to 0x0000(ES):0x9000(BX)
    mov dh, 5           ; from the boot disk.
    mov dl, [BOOT_DRIVE]
    call disk_load

    mov dx, [0x9000]     ; Print out the first loaded word, which
    call print_hex       ; we expect to be 0xdada, stored
                        ; at address 0x9000

    mov dx, [0x9000 + 512] ; Also, print the first word from the
    call print_hex        ; 2nd loaded sector: should be 0xface

    jmp $

%include "../print/print_string.asm" ; Re-use our print_string function
%include "../hex/print_hex.asm"      ; Re-use our print_hex function
%include "disk_load.asm"
; Include our new disk_load function

; Global variables
BOOT_DRIVE: db 0

; Bootsector padding
times 510-($-$$) db 0
dw 0xaa55

; We know that BIOS will load only the first 512-byte sector from the disk,
; so if we purposely add a few more sectors to our code by repeating some
; familiar numbers, we can prove to ourselves that we actually loaded those
; additional two sectors from the disk we booted from.
times 256 dw 0xdada
times 256 dw 0xface

```

## Chapter 4

---

# Entering 32-bit Protected Mode

---

It would be nice to continue working in the 16-bit real mode with which we have now become much better acquainted, but in order to make fuller use of the CPU, and to better understand how developments of CPU architectures can benefit modern operating systems, namely memory protection in hardware, then we must press on into 32-bit protected mode.

The main differences in 32-bit protected mode are:

- Registers are extended to 32 bits, with their full capacity being accessed by prefixing an **e** to the register name, for example: `mov ebx, 0x274fe8fe`
- For convenience, there are two additional general purpose segment registers, **fs** and **gs**.
- 32-bit memory offsets are available, so an offset can reference a whopping 4 GB of memory (`0xffffffff`).
- The CPU supports a more sophisticated --- though slightly more complex --- means of memory segmentation, which offers two big advantages:
  - Code in one segment can be prohibited from executing code in a more privileged segment, so you can protect your kernel code from user applications
  - The CPU can implement *virtual memory* for user processes, such that *pages* (i.e. fixed-sized chunks) of a process's memory can be swapped transparently between the disk and memory on an as-needed basis. This ensure main memory is used efficiently, in that code or data that is rarely executed needn't hog valuable memory.
- Interrupt handling is also more sophisticated.

[?]

The most difficult part about switching the CPU from 16-bit real mode into 32-bit protected mode is that we must prepare a complex data structure in memory called the *global descriptor table* (GDT), which defines memory segments and their protected-mode attributes. Once we have defined the GDT, we can use a special instruction to load it

into the CPU, before setting a single bit in a special CPU control register to make the actual switch.

This process would be easy enough if we didn't have to define the GDT in assembly language, but sadly this low-level switch-over is unavoidable if we later wish to load a kernel that has been compiled from a higher-level language such as C, which usually will be compiled to 32-bit instructions rather than the less-efficient 16-bit instructions.

Oh, there is one shocker that I nearly forgot to mention: we can no longer use BIOS once switched into 32-bit protected mode. If you thought making BIOS calls was low-level. This is like one step backwards, two steps forwards.

## 4.1 Adapting to Life Without BIOS

It is true: in our quest to make full use of the CPU, we must abandon all of those helpful routines provided by BIOS. As we will see when we look in more detail at the 32-bit protected mode switch-over, BIOS routines, having been coded to work only in 16-bit real mode, are no longer valid in 32-bit protected mode; indeed, attempting to use them would likely crash the machine.

So what this means is that a 32-bit operating system must provide its own drivers for all hardware of the machine (e.g. the keyboard, screen, disk drives, mouse, etc). Actually, it is possible for a 32-bit protected mode operating system to switch temporarily back into 16-bit mode whereupon it may utilise BIOS, but this technique can be more trouble than it is worth, especially in terms of performance.

The first problem we will encounter in switching to protected mode is knowing how to print a message on the screen, so we can see what is happening. Previously we have asked BIOS to print an ASCII character on the screen, but how did that result in the appropriate pixels being highlighted at the appropriate position of our computer's screen? For now, it suffices to know that the display device can be configured into one of several resolutions in one of two modes, *text mode* and *graphics mode*; and that what is displayed on the screen is a visual representation of a specific range of memory. So, in order to manipulate the screen, we must manipulate the specific memory range that it is using in its current mode. The display device is an example of memory-mapped hardware because it works in this way.

When most computers boot, despite that they may in fact have more advanced graphics hardware, they begin in a simple Video Graphics Array (VGA) colour text mode with dimensions 80x25 characters. In text mode, the programmer does not need to render individual pixels to describe specific characters, since a simple font is already defined in the internal memory of the VGA display device. Instead, each character cell of the screen is represented by two bytes in memory: the first byte is the ASCII code of the character to be displayed, and the second byte encodes the character's attributes, such as the foreground and background colour and if the character should be blinking.

So, if we'd like to display a character on the screen, then we need to set its ASCII code and attributes at the correct memory address for the current VGA mode, which is usually at address `0xb8000`. If we slightly modify our original (16-bit real mode) `print_string` routine so that it no longer uses the BIOS routine, we can create a 32-bit protected mode routine that writes directly to video memory, as in Figure 4.1.

Note that, although the screen is displayed as columns and rows, the video memory is simply sequential. For example, the address of the column 5 on row 3 can be calculated as follows: `0xb8000 + 2 * (row * 80 + col)`

```

[bits 32]
; Define some constants
VIDEO_MEMORY equ 0xb8000
WHITE_ON_BLACK equ 0x0f

; prints a null-terminated string pointed to by EDI
print_string_pm:
    pusha
    mov edi, VIDEO_MEMORY ; Set edi to the start of vid mem.

print_string_pm_loop:
    mov al, [ebx]          ; Store the char at EBX in AL
    mov ah, WHITE_ON_BLACK ; Store the attributes in AH

    cmp al, 0              ; if (al == 0), at end of string, so
    je done                ; jump to done

    mov [edi], ax          ; Store char and attributes at current
                           ; character cell.
    add ebx, 1             ; Increment EBX to the next char in string.
    add edi, 2             ; Move to next character cell in vid mem.

    jmp print_string_pm_loop ; loop around to print the next char.

print_string_pm_done :
    popa
    ret                    ; Return from the function

```

Figure 4.1: A routine for printing a string directly to video memory (i.e. without using BIOS).

The downside to our routine is that it always prints the string to the top-left of the screen, and so will overwrite previous messages rather than scrolling. We could spend time adding to the sophistication of this assembly routine, but let's not make things too hard for ourselves, since after we master the switch to protected mode, we will soon be booting code written in a higher level language, where we can make much lighter work of these things.

## 4.2 Understanding the Global Descriptor Table

It is important to understand the main point of this GDT, that is so fundamental to the operation of protected mode, before we delve into the details. Recall from Section XXX that the design rationale of segment-based addressing in the classical 16-bit real mode was to allow the programmer to access (albeit slightly, by today's standards) more memory than a 16-bit offset would allow. As an example of this, suppose that the programmer wanted to store the value of `ax` at the address `0x4fe56`. Without segment-

based addressing, the best the programmer could do is this:

```
mov [0xffff], ax
```

which falls way short of the intended address. Whereby, using a segment register, the task could be achieved as follows:

```
mov bx, 0x4000
mov es, bx
mov [es:0xfe56], ax
```

Although the general idea of segmenting memory and using offsets to reach into those segments has remained the same, the way that it is implemented in protected mode has completely changed, primarily to afford more flexibility. Once the CPU has been switched into 32-bit protected mode, the process by which it translates logical addresses (i.e. the combination of a segment register and an offset) to physical address is completely different: rather than multiply the value of a segment register by 16 and then add to it the offset, a segment register becomes an index to a particular *segment descriptor* (SD) in the GDT.

A segment descriptor is an 8-byte structure that defines the following properties of a protected-mode segment:

- Base address (32 bits), which defines where the segment begins in physical memory
- Segment Limit (20 bits), which defines the size of the segment
- Various flags, which affect how the CPU interprets the segment, such as the privilege level of code that runs within it or whether it is read- or write-only.

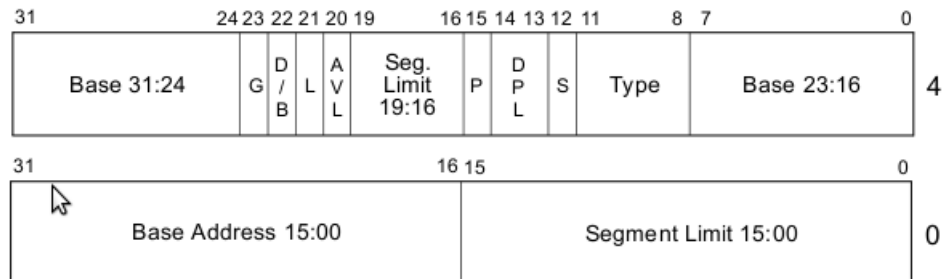
Figure 4.2 shows the actual structure of the segment descriptor. Notice how, just to add to the confusion, the structure fragments the base address and segment limit throughout the structure, so, for example, the lower 16 bits of the segment limit are in the first two bytes of the structure but the higher 4-bits are at the start of the seventh byte of the structure. Perhaps this was done as some kind of joke, or more likely it has historic roots or was influenced by the CPU's hardware design.

We will not concern ourselves with details of all of the possible configurations of segment descriptors, a full explanation of which is given in Intel's Developer Manual [?], but we will learn what we have to in order to get our code running in 32-bit protected mode.

The simplest workable configuration of segment registers is described by Intel as the *basic flat model*, whereby two overlapping segments are defined that cover the full 4 GB of addressable memory, one for *code* and the other for *data*. The fact that in this model these two segments overlap means that there is no attempt to protect one segment from the other, nor is there any attempt to use the paging features for virtual memory. It pays to keep things simple early on, especially since later we may alter the segment descriptors more easily once we have booted into a higher-level language.

In addition to the *code* and *data* segments, the CPU requires that the first entry in the GDT purposely be an invalid *null descriptor* (i.e. a structure of 8 zero bytes). The null descriptor is a simple mechanism to catch mistakes where we forget to set a particular segment register before accessing an address, which is easily done if we had some segment registers set to 0x0 and forgot to update them to the appropriate segment descriptors after switching to protected mode. If an addressing attempt is made with the null descriptor, then the CPU will raise an exception, which essentially is an interrupt ---

and which, although not too dissimilar as a concept, is not to be confused with exceptions in higher level languages such as Java.



- L — 64-bit code segment (IA-32e mode only)
- AVL — Available for use by system software
- BASE — Segment base address
- D/B — Default operation size (0 = 16-bit segment; 1 = 32-bit segment)
- DPL — Descriptor privilege level
- G — Granularity
- LIMIT — Segment Limit
- P — Segment present
- S — Descriptor type (0 = system; 1 = code or data)
- TYPE — Segment type

Figure 4.2: Segment descriptor structure.

Our *code* segment will have the following configuration:

- Base: 0x0
- Limit: 0xfffff
- Present: 1, since segment is present in memory - used for virtual memory
- Privilege: 0, ring 0 is the highest privilege
- Descriptor type: 1 for code or data segment, 0 is used for traps
- Type:
  - Code: 1 for code, since this is a code segment
  - Conforming: 0, by not conforming it means code in a segment with a lower privilege may not call code in this segment - this a key to memory protection
  - Readable: 1, 1 if readable, 0 if execute-only. Readable allows us to read constants defined in the code.
  - Accessed: 0 This is often used for debugging and virtual memory techniques, since the CPU sets the bit when it accesses the segment
- Other flags

- Granularity: 1, if set, this multiplies our limit by 4 K (i.e.  $16 \times 16 \times 16$ ), so our 0xffff would become 0xffff000 (i.e. shift 3 hex digits to the left), allowing our segment to span 4 Gb of memory
- 32-bit default: 1, since our segment will hold 32-bit code, otherwise we'd use 0 for 16-bit code. This actually sets the default data unit size for operations (e.g. push 0x4 would expand to a 32-bit number ,etc.)
- 64-bit code segment: 0, unused on 32-bit processor
- AVL: 0, We can set this for our own uses (e.g. debugging) but we will not use it

Since we are using a simple flat model, with two overlapping code and data segments, the *data* segment will be identical but for the type flags:

- Code: 0 for data
- Expand down: 0 . This allows the segment to expand down - TODO explain this
- Writable: 1. This allows the data segment to be written to, otherwise it would be read only
- Accessed: 0 This is often used for debugging and virtual memory techniques, since the CPU sets the bit when it accesses the segment

Now that we have seen an actual configuration of two segments, exploring most of the possible segment descriptor settings, it should be clearer how protected mode offers much more flexibility in the partitioning of memory than real mode.

### 4.3 Defining the GDT in Assembly

Now that we understand what segment descriptors to include in our GDT for the basic flat model, let us look at how we might actually represent the GDT in assembly, a task that requires more patience than anything else. Whilst you're experiencing the sheer tediousness of this, keep in mind the significance of it: what we do here will allow us soon to boot our operating system kernel, which we will write in a higher level language, then --- for want of a better quote --- our small steps will turn into giant leaps.

We have already seen examples of how to define data within our assembly code, using the `db`, `dw`, and `dd` assembly directives, and these are exactly what we must use to put in place the appropriate bytes in the segment descriptor entries of our GDT.

Actually, for the simple reason that the CPU needs to know how long our GDT is, we don't actually directly give the CPU the start address of our GDT but instead give it the address of a much simpler structure called the GDT descriptor (i.e. something that describes the GDT). The GDT is a 6-byte structure containing:

- GDT size (16 bits)
- GDT address (32 bits)

Note, when working in such a low-level language with complex data structures like these, we cannot add *enough* comments. The following code defines our GDT and the GDT descriptor; in the code, notice how we use `db`, `dw`, etc. to fill out parts of the

structure and how the flags are more conveniently defined using literal binary numbers, that are suffixed with **b**:

```
; GDT
gdt_start:

gdt_null: ; the mandatory null descriptor
    dd 0x0 ; 'dd' means define double word (i.e. 4 bytes)
    dd 0x0

gdt_code: ; the code segment descriptor
    ; base=0x0, limit=0xffff,
    ; 1st flags: (present)1 (privilege)00 (descriptor type)1 -> 1001b
    ; type flags: (code)1 (conforming)0 (readable)1 (accessed)0 -> 1010b
    ; 2nd flags: (granularity)1 (32-bit default)1 (64-bit seg)0 (AVL)0 -> 1100b
    dw 0xffff ; Limit (bits 0-15)
    dw 0x0 ; Base (bits 0-15)
    db 0x0 ; Base (bits 16-23)
    db 10011010b ; 1st flags, type flags
    db 11001111b ; 2nd flags, Limit (bits 16-19)
    db 0x0 ; Base (bits 24-31)

gdt_data: ; the data segment descriptor
    ; Same as code segment except for the type flags:
    ; type flags: (code)0 (expand down)0 (writable)1 (accessed)0 -> 0010b
    dw 0xffff ; Limit (bits 0-15)
    dw 0x0 ; Base (bits 0-15)
    db 0x0 ; Base (bits 16-23)
    db 10010010b ; 1st flags, type flags
    db 11001111b ; 2nd flags, Limit (bits 16-19)
    db 0x0 ; Base (bits 24-31)

gdt_end: ; The reason for putting a label at the end of the
        ; GDT is so we can have the assembler calculate
        ; the size of the GDT for the GDT descriptor (below)

; GDT descriptor
gdt_descriptor:
    dw gdt_end - gdt_start - 1 ; Size of our GDT, always less one
                                ; of the true size
    dd gdt_start ; Start address of our GDT

; Define some handy constants for the GDT segment descriptor offsets, which
; are what segment registers must contain when in protected mode. For example,
; when we set DS = 0x10 in PM, the CPU knows that we mean it to use the
; segment described at offset 0x10 (i.e. 16 bytes) in our GDT, which in our
; case is the DATA segment (0x0 -> NULL; 0x08 -> CODE; 0x10 -> DATA)
CODE_SEG equ gdt_code - gdt_start
DATA_SEG equ gdt_data - gdt_start
```

## 4.4 Making the Switch

Once both the GDT and the GDT descriptor have been prepared within our boot sector, we are ready to instruct the CPU to switch from 16-bit real mode into 32-bit protected mode.

Like I said before, the actual switchover is fairly straight forward to code, but it is important to understand the significance of the steps involved.

The first thing we have to do is disable interrupts using the `cli` (clear interrupt) instruction, which means the CPU will simply ignore any future interrupts that may happen, at least until interrupts are later enabled. This is very important, because, like segment based addressing, interrupt handling is implemented completely differently in protected mode than in real mode, making the current IVT that BIOS set up at the start of memory completely meaningless; and even if the CPU could map interrupt signals to their correct BIOS routines (e.g. when the user pressed a key, store its value in a buffer), the BIOS routines would be executing 16-bit code, which will have no concept of the 32-bit segments we defined in our GDT and so will ultimately crash the CPU by having segment register values that assume the 16-bit real mode segmenting scheme.

The next step is to tell the CPU about the GDT that we just prepared --- with great pain. We use a single instruction to do this, to which we pass the GDT descriptor:

```
lgdt [gdt_descriptor]
```

Now that all is in-place, we make the actual switch over, by setting the first bit of a special CPU control register, `cr0`. Now, we cannot set that bit directly on the register, so we must load it into a general purpose register, set the bit, then store it back into `cr0`. Similarly to how we used the `and` instruction in Section XXX to exclude bits from a value, we can use the `or` instruction to include certain bits into a value (i.e. without disturbing any other bits that, for some important reason, may have been set already in the control register) [?].

```
mov eax, cr0    ; To make the switch to protected mode, we set
or  eax, 0x1    ; the first bit of CR0, a control register
mov cr0, eax    ; Update the control register
```

After `cr0` has been updated, the CPU is in 32-bit protected mode [?].

That last statement is not entirely true, since modern processors use a technique called *pipelining*, that allows them to process different stages of an instruction's execution in parallel (and I am talking about single CPUs as opposed to parallel CPUs), and therefore in less time. For example, each instruction might be *fetched* from memory, *decoded* into microcode instructions, *executed*, then perhaps the result is *stored* back to memory; and since these stages are semi-independent, they could all be done within the same CPU cycle but within different circuitry (e.g. the previous instruction could be *decoded* whilst the next is *fetched*) [?].

We do not normally need to worry about CPU internals such as pipelining when programming the CPU, but switching CPU modes is a special case, since there is a risk that the CPU may process some stages of an instruction's execution in the wrong mode. So what we need to do, immediately after instructing the CPU to switch mode, is to force the CPU to finish any jobs in its pipeline, so that we can be confident that all future instructions will be executed in the correct mode.

Now, pipelining works very well when the CPU knows about the next few instructions that will be coming over the horizon, since it can pre-fetch them, but it doesn't like instructions such as `jmp` or `call`, because until those instructions have been executed fully the CPU can have no idea about the instructions that will follow them, especially if we use a *far* jump or call, which means that we jump to another segment. So immediately after instructing the CPU to switch mode, we can issue a far jump, which will force the

CPU to flush the pipeline (i.e. complete all of instructions currently in different stages of the pipeline).

To issue a far jump, as opposed to a near (i.e. standard) jump, we additionally provide the target segment, as follows:

```
jmp <segment>:<address offset>
```

For this jump, we need to think carefully about where we wish to land. Suppose we set up a label in our code such as `start_protected_mode` that marks the beginning of our 32-bit code. As we have just discussed, a *near* jump, such as `jmp start_protected_mode` may not be sufficient to flush the pipeline, and, besides we are now in some strange limbo, since our current code segment (i.e. `cs`) will not be valid in protected mode. So, we must update our `cs` register to the offset of the code segment descriptor of our GDT. Since the segment descriptors are each 8 bytes long, and since our code descriptor was the second descriptor in our GDT (the null descriptor was the first), its offset will be `0x8`, and so this value is what we must now set our code segment register to. Note that, by the very definition of a far jump, it will automatically cause the CPU to update our `cs` register to the target segment. Making handy use of labels, we got our assembler to calculate these segment descriptor offsets and store them as the constants `CODE_SEG` and `DATA_SEG`, so now we arrive at our jump instruction:

```
jmp CODE_SEG:start_protected_mode

[bits 32]

start_protected_mode:
...           ; By now we are assuredly in 32-bit protected mode.
...
```

Note that, in fact, we don't need to jump very far at all in terms of the physical distance between the where we jumped from and where we landed, but the importance was in how we jump.

Note also that we need to use the `[bits 32]` directive to tell our assembler that, from that point onwards, it should encode in 32-bit mode instructions. Note, though, that this does not mean we cannot use 32-bit instructions in 16-bit real mode, just that the assembler must encode those instructions slightly differently than in 32-bit protected mode [?]. Indeed, when switching to protected mode, we made use of the 32-bit register `eax` to set the control bit.

Now we are in 32-bit protected mode. A good thing to do once we have entered 32-bit mode proper is to update all of the other segment registers so they now point to our 32-bit data segment (rather than the now-invalid real mode segments) and update the position of the stack.

We can combine the whole process into a re-usable routine, as in Figure XXX.

```
[bits 16]
; Switch to protected mode
switch_to_pm:

cli           ; We must switch off interrupts until we have
              ; set-up the protected mode interrupt vector
              ; otherwise interrupts will run riot.
```

```

    lgdt [gdt_descriptor] ; Load our global descriptor table, which defines
                           ; the protected mode segments (e.g. for code and data)

    mov eax, cr0           ; To make the switch to protected mode, we set
    or  eax, 0x1          ; the first bit of CR0, a control register
    mov  cr0, eax

    jmp CODE_SEG:init_pm   ; Make a far jump (i.e. to a new segment) to our 32-bit
                           ; code. This also forces the CPU to flush its cache of
                           ; pre-fetched and real-mode decoded instructions, which can
                           ; cause problems.

[bits 32]
; Initialise registers and the stack once in PM.
init_pm:

    mov ax, DATA_SEG     ; Now in PM, our old segments are meaningless,
    mov ds, ax            ; so we point our segment registers to the
    mov ss, ax            ; data selector we defined in our GDT
    mov es, ax
    mov fs, ax
    mov gs, ax

    mov ebp, 0x90000      ; Update our stack position so it is right
    mov esp, ebp          ; at the top of the free space.

    call BEGIN_PM         ; Finally, call some well-known label

```

## 4.5 Putting it all Together

Finally, we can include all of our routines into a boot sector that demonstrates the switch from 16-bit real mode into 32-bit protected mode.

```

; A boot sector that enters 32-bit protected mode.
[org 0x7c00]

    mov bp, 0x9000        ; Set the stack.
    mov sp, bp

    mov bx, MSG_REAL_MODE
    call print_string

    call switch_to_pm     ; Note that we never return from here.

    jmp $

%include "../print/print_string.asm"
%include "gdt.asm"
%include "print_string_pm.asm"
%include "switch_to_pm.asm"

[bits 32]

```

```
; This is where we arrive after switching to and initialising protected mode.
BEGIN_PM:

    mov ebx, MSG_PROT_MODE
    call print_string_pm    ; Use our 32-bit print routine.

    jmp $                  ; Hang.

; Global variables
MSG_REAL_MODE    db "Started in 16-bit Real Mode", 0
MSG_PROT_MODE    db "Successfully landed in 32-bit Protected Mode", 0

; Bootsector padding
times 510-($-$$) db 0
dw 0xaa55
```

## Chapter 5

---

# Writing, Building, and Loading Your Kernel

---

So far, we have learnt a lot about how the computer really works, by communicating with it in the low-level assembly language, but we've also seen how it can be very slow to progress in such a language: we need to think very carefully even about the simplest of control structures (e.g. `if (<something>) <do this> else <do that>`), and we have to worry about how best to make use of the limited number of registers, and juggle with the stack. Another drawback of us continuing in assembly language is that it is closely tied to the specific CPU architecture, and so it would be harder for us to port our operating system to another CPU architecture (e.g. ARM, RISC, PowerPC).

Luckily, other programmers also got fed up of writing in assembly, so decided to write higher-level language compilers (e.g. FORTRAN, C, Pascal, BASIC, etc.), that would transform more intuitive code into assembly language. The idea of these compilers is to map higher level constructs, such as control structures and function calls onto assembly template code, and so the downside --- and there usually always is a downside --- is that the generic templates may not always be optimal for specific functionality. Without further ado, let us look at how C code is transformed into assembly to demystify the role of the compiler.

## 5.1 Understanding C Compilation

Let's write some small code snippets in C and see what sort of assembly code they generate. This is a great way of learning about how C works.

### 5.1.1 Generating Raw Machine Code

```
// Define an empty function that returns an integer
```

```
int my_function() {
    return 0xbaba;
}
```

Save the code in Figure XXXX into a file called `basic.c`, and compile it as follows:

```
$gcc -ffreestanding -c basic.c -o basic.o
```

This will produce an *object file*, which, being completely unrelated, is not to be confused with the concept of object-oriented programming. Rather than compiling directly into machine code, the compiler outputs *annotated* machine code, where meta information, such as textual labels, that are redundant for execution, remain present to enable more flexibility in how the code is eventually put together. One big advantage of this intermediary format is that the code may be easily relocated into a larger binary file when linked in with routines from other routines in other libraries, since code in the object file uses relative rather than absolute internal memory references. We can get a good view of an object file's contents with the following command:

```
$objdump -d basic.o
```

The output of this command will give something like that in Figure XXX. Note that we can see some assembly instructions and some additional details about the code. Note that the syntax of the assembly is very slightly different to that used by `nasm`, so simply ignore this part, since we will soon see it in a more familiar format.

```
basic.o:          file format elf32-i386

Disassembly of section .text:

00000000 <my_function>:
   0: 55                push    %ebp
   1: 89 e5             mov     %esp,%ebp
   3: b8 ba ba 00 00    mov     $0xbaba,%eax
   8: 5d                pop     %ebp
   9: c3                ret
```

In order to create the actual executable code (i.e. that will run on our CPU), we have to use a *linker*, whose role is to link together all of the routines described in the input object files into one executable binary file, effectively stitching them together and converting those relative addresses into absolute addresses within the aggregated machine code, for example: `call <function_X_label>` will become `call 0x12345`, where `0x12345` is the offset within the output file that the linker decided to place the code for the routine denoted by `function_X_label`.

In our case, though, we do not want to link with any routines from any other object files --- we will look at this shortly --- but nevertheless the linker will convert our annotated machine code file into a raw machine code file. To output raw machine code into a file `basic.bin`, we can use the following command:

```
$ld -o basic.bin -Ttext 0x0 --oformat binary basic.o
```

Note that, like the compiler, the linker can output executable files in various formats, some of which may retain meta data from the input object files. This is useful for executables that are hosted by an operating system, such as the majority of programs we will write on a platform such as Linux or Windows, since meta data can be retained to describe how those applications are to be loaded into memory; and for debugging purposes, for example: the information that a process crashed at instruction address `0x12345678` is far less useful to a programmer than information presented using redundant, non-executable meta-data that a process crashed in function `my_function`, file `basic.c`, on line 3.

Anyhow, since we are interested in writing an operating system, it would be no good trying to run machine code intermingled with meta data on our CPU, since unaware the CPU will execute every byte as machine code. This is why we specify an output format of (raw) `binary`.

The other option we used was `-Ttext 0x0`, which works in the same way as the `org` directive we used in our earlier assembly routines, by allowing us to tell the compiler to offset label addresses in our code (e.g. for any data we specify in the code, such as strings like `'Hello, World'`) to their absolute memory addresses when later loaded to a specific origin in memory. For now this is not important, but when we come to load kernel code into memory, it is important that we set this to the address we plan to load to.

Now we have successfully compiled the C code into a raw machine code file, that we could (once we have figured out how to load it) run on our CPU, so let's see what it looks like. Luckily, since assembly maps very closely to machine code instructions, if you are ever given a file containing only machine code, you can easily disassemble it to view it in assembly. Ah, yes; this is another benefit of understanding a little of assembly, because you can potentially reverse-engineer any software that lands on your lap minus the original source code, even more successfully if the developer left in some meta data for you --- which they nearly always do. The only problem with disassembling machine code is that some of those bytes may have been reserved as data but will show up as assembly instructions, though in our simple C program we didn't declare any data. To see what machine code the compiler actually generated from our C source code, run the following command:

```
$ndisasm -b 32 basic.bin > basic.dis
```

The `-b 32` simply tells the disassembler to decode to 32-bit assembly instructions, which is what our compiler generates. Figure XXX shows the assembly code generated by gcc for our simple C program.

|          |            |                             |
|----------|------------|-----------------------------|
| 00000000 | 55         | <code>push ebp</code>       |
| 00000001 | 89E5       | <code>mov ebp,esp</code>    |
| 00000003 | B8BABA0000 | <code>mov eax,0xbaba</code> |
| 00000008 | 5D         | <code>pop ebp</code>        |
| 00000009 | C3         | <code>ret</code>            |

So here it is: gcc generated some assembly code not too dissimilar to that which we have been writing ourselves already. The three columns output from the disassembler,

from left to right, show the file offsets of the instructions, the machine code, and the equivalent assembly instructions. Although our function does a very simple thing, there is some additional code in there that seems to be manipulating the stack's base and top registers, `ebp` and `esp`. C makes heavy use of the stack for storing variables that are local to a function (i.e. variables that are no-longer needed when the function returns), so upon entering a function, the stack's base pointer (`ebp`) is increased to the current top of the stack (`mov ebp, esp`), effectively creating a local, initially empty stack above the stack of the function that called our function. This process is often referred to as the function setting up its *stack frame*, in which it will allocate any local variables. However, if prior to returning from our function we failed to restore the stack frame to that originally set up by our caller, the calling function would get in a real mess when trying to access its local variables; so before updating the base pointer for our stack frame, we must store it, and there is no better place to store it than the top of the stack (`push ebp`).

After preparing our stack frame, which, sadly, doesn't actually get used in our simple function, we see how the compiler handles the line `return 0xbaba;`: the value `0xbaba` is stored in the 32-bit register `eax`, which is where the calling function (if there were one) would expect to find the returned value, similarly to how we had our own convention of using certain registers to pass arguments to our earlier assembly routines, for example: our `print_string` routine expected to find the address of the string to be printed in the register `bx`.

Finally, before issuing `ret` to return to the caller, the function pops the original stack base pointer off the stack (`pop ebp`), so the calling function will be unaware that its own stack frame was ever changed by the called function. Note that we didn't actually change the top of the stack (`esp`), since in this case our stack frame was used to store nothing, so the untouched `esp` register did not require restoring.

Now we have a good idea about how C code translates into assembly, so let's prod the compiler a little further until we have sufficient understanding to write a simple kernel in C.

[?]

### 5.1.2 Local Variables

Now write the code in Figure XXX into a file called `local_var.c` and compile, link, and disassemble it as before.

```
// Declare a local variable.
int my_function() {
    int my_var = 0xbaba;
    return my_var;
}
```

Now the compiler generates the assembly code in Figure XXX.

```

00000000  55                push ebp
00000001  89E5             mov ebp,esp
00000003  83EC10          sub esp,byte +0x10
00000006  C745FCBABA0000  mov dword [ebp-0x4],0xbaba
0000000D  8B45FC          mov eax,[ebp-0x4]
00000010  C9              leave
00000011  C3              ret

```

The only difference now is that we actually allocate a local variable, `my_var`, but this provokes an interesting response from the compiler. As before, the stack frame is established, but then we see `sub esp, byte +0x10`, which is subtracting 16 (`0x10`) bytes from the top of the stack. Firstly, we have to (constantly) remind ourselves that the stack grows downwards in terms of memory addresses, so in simpler terms this instructions means, 'allocate another 16 bytes on the top of stack'. We are storing an `int`, which is a 4-byte (32-bit) data type, so why have 16 bytes been allocated on the stack for this variable, and why not use `push`, which allocates new stack space automatically? The reason the compiler manipulates the stack in this way is one of optimisation, since CPUs often operate less efficiently on a datatype that is not aligned on memory boundaries that are multiples of the datatype's size [?]. Since C would like all variables to be properly aligned, it uses the maximum datatype width (i.e. 16 bytes) for all stack elements, at the cost of wasting some memory.

The next instruction, `mov dword [ebp-0x4],0xbaba`, actually stores our variable's value in the newly allocated space on the stack, but without using `push`, for the previously given reason of stack efficiency (i.e. the size of the datatype stored is less than the stack space reserved). We understand the general use of the `mov` instruction, but two things that deserve some explanation here are the use of `dword` and `[ebp-0x4]`:

- `dword` states explicitly that we are storing a *double word* (i.e. 4 bytes) on the stack, which is the size of our `int` datatype. So the actual bytes stored would be `0x0000baba`, but without being explicit could easily be `0xbaba` (i.e. 2 bytes) or `0x000000000000baba` (i.e. 8 bytes), which, although the same value, have different ranges.
- `[ebp-0x4]` is an example of a modern CPU shortcut called *effective address computation* [?], which is more impressive that the assembly code seems to reflect. This part of the instruction references an address that is calculated *on-the-fly* by the CPU, based on the current address of register `ebp`. At a glance, we might think our assembler is manipulating a constant value, as it would if we wrote something like this `mov ax, 0x5000 + 0x20`, where our assembler would simply pre-process this into `mov ax, 0x5020`. But only once the code is run would the value of any register be known, so this definitely is not pre-processing; it forms a part of the CPU instruction. With this form of addressing the CPU is allowing us to do more per instruction cycle, and is good example of how CPU hardware has adapted to better suit programmers. We could write the equivalent, without such address manipulation, less efficiently in the following three lines of code:

```

mov eax, ebp      ; EAX = EBP
sub eax, 0x4      ; EAX = EAX - 0x4
mov [eax], 0xbaba ; store 0xbaba at address EAX

```

So the value `0xbaba` is stored directly to the appropriate position of the stack, such that it will occupy the first 4 bytes above (though physically below, since the stack grows downwards) the base pointer.

Now, being a computer program, our compiler can distinguish different numbers as easily as we can distinguish different variable names, so what we think of as the variable `my_var`, the compiler will think of as the address `ebp-0x4` (i.e. the first 4 bytes of the stack). We see this in the next instruction, `mov eax,[ebp-0x4]`, which basically means, 'store the contents of `my_var` in `eax`', again using efficiently address computation; and we know from the previous function that `eax` is used to return a variable to the caller of our function.

Now, before the `ret` instruction, we see something new: the `leave` instruction. Actually, the `leave` instruction is an alternative to the following steps, that restore the original stack of the caller, reciprocal of the first two instruction of the function:

```
mov esp, ebp ; Put the stack back to as we found it.
pop ebp
```

Though only a single instruction, it is not always the case that `leave` is more efficient than the separate instructions [?]. Since our compiler chose to use this instruction, we will leave that particular discussion to other people.

### 5.1.3 Calling Functions

Not let's look at the C code in Figure XXX, which has two functions, where one function, `caller_function`, calls the other, `callee_function`, passing it an integer argument. The called function simply returns the argument it was passed.

```
void caller_function() {
    callee_function(0xdede);
}

int callee_function(int my_arg) {
    return my_arg;
}
```

If we compile and disassemble the C code, we will get something similar to that in Figure XXX.

|          |                |                                     |
|----------|----------------|-------------------------------------|
| 00000000 | 55             | <code>push ebp</code>               |
| 00000001 | 89E5           | <code>mov ebp,esp</code>            |
| 00000003 | 83EC08         | <code>sub esp,byte +0x8</code>      |
| 00000006 | C70424DEDE0000 | <code>mov dword [esp],0xdede</code> |
| 0000000D | E802000000     | <code>call dword 0x14</code>        |
| 00000012 | C9             | <code>leave</code>                  |
| 00000013 | C3             | <code>ret</code>                    |
| 00000014 | 55             | <code>push ebp</code>               |
| 00000015 | 89E5           | <code>mov ebp,esp</code>            |
| 00000017 | 8B4508         | <code>mov eax,[ebp+0x8]</code>      |

|          |    |         |
|----------|----|---------|
| 0000001A | 5D | pop ebp |
| 0000001B | C3 | ret     |

Firstly, notice how we can differentiate between assembly code of the two functions by looking for the tell-tale `ret` instruction that always appears as the last instruction of a function. Next, notice how the upper function uses the assembly instruction `call`, which we know is used to jump to another routine from which usually we expect to return. This must be our `caller_function`, that is calling `callee_function` at offset `0x14` of the machine code. The most interesting lines here are those immediately before the call, since they are somehow ensuring that the argument `my_arg` is passed to `callee_function`. After establishing its stack frame, as we have seen to be common to all functions, `caller_function` allocates 8 bytes on the top of the stack (`sub esp, byte +0x8`), then stores our passed value, `0xdede`, into that stack space (`mov dword [esp], 0xdede`).

So let's see how `callee_function` accesses that argument. From offset `0x14`, we see that `callee_function` establishes its stack frame as usual, but then look at what it stores in the `eax` register, a register that we know from our earlier analysis is used to hold a function's return value: it stores the contents of address `[ebp + 0x8]`. Here we have to remind ourselves again of that confusing fact that the stack grows downwards in memory, so in terms of logically-more-sensible upward growing stack, `ebp + 0x8` is 8 bytes *below* our stack's base, so we are actually reaching into the stack frame of the function that called us to get the argument's value. This is what we'd expect, of course, because the caller put that value onto the top of their stack, then we put our stack base at the top of their stack to establish our stack frame.

It is very useful to know the calling convention used by any high-level language compiler when interfacing its code in assembly. For example, the default calling convention of C is to push arguments onto the stack in reverse order, so the first argument is on the top of the stack. To mix up the order of arguments would certainly cause the program to perform incorrectly, and likely crash.

### 5.1.4 Pointers, Addresses, and Data

When working in a high-level language we can easily find ourselves forgetting about the fact that variables are simply references to allocated memory addresses, where sufficient space has been reserved to accommodate their particular data type. This is because, in most cases when we are dealing with variables, we are really only interested in the values that they hold, rather than where they reside in memory. Consider the following snippet of C code:

```
int a = 3;
int b = 4;
int total = a + b;
```

Now that we have more of an awareness about how the computer will actually perform these simple C instructions, we could make a well informed assumption that the instruction `int a = 3;` will involve two main steps: firstly, at least 4 bytes (32 bits) will be reserved, perhaps on the stack, to hold the value; then, the value `3` will be stored at the reserved address. The same would be the case for the second line. And in the line `int`

`total = a + b;`, some more space will be reserved for the variable `total`, and in it will be stored the addition of the *contents of addresses* pointed to by the labels `a` and `b`.

Now, suppose that we'd like to store a value at a specific address of memory; for example, like we have done in assembly, to write characters directly to the video memory at address `0xb8000` when BIOS was no longer available. How would we do that in C, when it seems that any value we wish to store must be in an address that has been determined by the compiler? Indeed, some high-level languages do not allow us to access memory in this way at all, which essentially is breaking the fluffy abstraction of the language. Luckily, C allows us to use *pointer* variables, that are datatypes used for storing addresses (rather than values), and which we can *dereference* to read or write data to wherever they point.

Now, technically, all pointer variables are the same datatype (e.g. a 32-bit memory address), but usually we plan to read and write specific datatypes from and to the address pointed to by a pointer, so we tell the compiler that, say, *this* is a pointer to a `char` and *that* is a pointer to an `int`. This is really a convenience, so that we do not always have to tell the compiler how many bytes it should read and write from the address held in a certain pointer. The syntax for defining and using pointers is shown in Figure XXX.

```
// Here, the star following the type means that this is not a variable to hold
// a char (i.e. a single byte) but a pointer to the ADDRESS of a char,
// which, being an address, will actually require the allocation of at least
// 32 bits.
char* video_address = 0xb8000;

// If we'd like to store a character at the address pointed to, we make the
// assignment with a star-prefixed pointer variable. This is known as
// dereferencing a pointer, because we are not changing the address held by
// the pointer variable but the contents of that address.
*video_address = 'X';

// Just to emphasise the purpose of the star, an omission of it, such as:
video_address = 'X';
// would erroneously store the ASCII code of 'X' in the pointer variable,
// such that it may later be interpreted as an address.
```

In C code we often see `char*` variables used for strings. Let's think about why this is. If we'd like to store a single `int` or `char`, then we know that they are both fixed sized datatypes (i.e. we know how many bytes they will use), but a string is an *array* of datatypes (usually of `char`), which may be of any length. So, since a single datatype cannot hold an entire string, only one element of it, we can use a pointer to a `char`, and set it to the memory address of the first character of the string. This is actually what we did in our assembly routines, such as `print_string`, where we allocated a string of characters (e.g. `'Hello, World'`) somewhere within our code, then, to print a particular string, we passed the first character's address via the `bx` register.

Let's look at an example of what the compiler does when we set up a string variable. In Figure XXX, we define a simple function that does nothing else other than allocate a string to a variable.

```
void my_function() {
    char* my_string = "Hello";
}
```

As before, we can disassemble to give something like that in Figure XXX.

|          |                |                                |
|----------|----------------|--------------------------------|
| 00000000 | 55             | push ebp                       |
| 00000001 | 89E5           | mov ebp,esp                    |
| 00000003 | 83EC10         | sub esp,byte +0x10             |
| 00000006 | C745FA48656C6C | mov dword [ebp-0x6],0x6c6c6548 |
| 0000000D | 66C745FE6F00   | mov word [ebp-0x2],0x6f        |
| 00000013 | C9             | leave                          |
| 00000014 | C3             | ret                            |

Firstly, to get our bearings we look for the `ret` instruction, that marks the end of the function. We see that the first two instructions of the function set the stack frame up, as usual. The next instruction, which we have also seen before, `sub esp,byte +0x10`, allocates 16 bytes on the stack to store our local variable. Now, the next instruction, `mov dword [ebp-0x4],0xf`, should have a familiar form, since it stores a value in our variable; but why does it store the number `0xf` --- we didn't tell it to do that, did we? After storing this suspicious value, we see the function politely revert the stack to the callers stack frame (`leave`) then return (`ret`). But look, there are five more instructions after the end of the function! What do you think the instruction `dec eax` is doing? Perhaps it decreases the value of `eax` by 1, but why? And what about the rest of the instructions?

At times like this we need to do a sanity check, and remember that: the disassembler cannot distinguish between code and data; and somewhere in that code must be data for the string we defined. Now, we know that our function consists of the first half of the code, since these instructions made sense to us, and they ended with `ret`. If we now assume that the rest of the code is in fact our data, then the suspicious value, `0xf`, that was stored in our variable makes sense, because it is the offset from the start of the code to where the data begins: our pointer variable is being set the the address of the data. To reassure our instincts, if we looked up in an ASCII table the character values of our string `'Hello'`, we would find them to be `0x48`, `0x65`, `0x6c`, `0x6c`, and `0x6f`. Now it is becoming clear, because if we look at the middle column of the disassembler output we see that these are the machine code bytes for those strange instructions that didn't seem to make sense; we see also that the very last byte is `0x0`, which C adds automatically to the end of strings, so that, like in our assembly routine `print_string`, during processing we can easily determine when we reach the end of the string.

## 5.2 Executing our Kernel Code

Enough of the theory, let's boot and execute the simplest of kernels written in C. This step will use all we have learnt so far, and will pave the way to faster progress in developing our operating system's features.

The involved steps are as follows:

- Write and compile the kernel code.
- Write and assemble the boot sector code
- Create a kernel image that includes not only our boot sector but our compiled kernel code
- Load our kernel code into memory
- Switch to 32-bit protected mode
- Begin executing our kernel code

### 5.2.1 Writing our Kernel

This will not take long, since, for the moment, the main function of our kernel nearly is to let us know it has been successfully loaded and executed. We can elaborate on the kernel later, so it is important initially to keep things simple. Save the code in Figure XXX into a file called `kernel.c`.

```
void main() {  
    // Create a pointer to a char, and point it to the first text cell of  
    // video memory (i.e. the top-left of the screen)  
    char* video_memory = (char*) 0xb8000;  
    // At the address pointed to by video_memory, store the character 'X'  
    // (i.e. display 'X' in the top-left of the screen).  
    *video_memory = 'X';  
}
```

Compile this to raw binary as follows:

```
$gcc -ffreestanding -c kernel.c -o kernel.o  
$ld -o kernel.bin -Ttext 0x1000 kernel.o --oformat binary
```

Note that, now, we tell the linker that the origin of our code once we load it into memory will be `0x1000`, so it knows to offset local address references from this origin, just like we use `[org 0x7c00]` in our boot sector, because that is where BIOS loads and then begins to execute it.

### 5.2.2 Creating a Boot Sector to Bootstrap our Kernel

We are going to write a boot sector now, that must bootstrap (i.e. load and begin executing) our kernel from the disk. Since the kernel was compiled as 32-bit instructions, we are going to have to switch into 32-bit protected mode before executing the kernel code. We know that BIOS will load only our boot sector (i.e the first 512 bytes of the disk), and so not our kernel, when the computer boots, but in Section XXX we have seen how we can use the BIOS disk routines to have our boot sector load additional sectors from a disk, and we are vaguely aware that, after we switch into protected mode, the

lack of BIOS will make it hard for us to use the disk: we would have to write a floppy or hard disk driver ourselves!

To simplify the problem of which disk and from which sectors to load the kernel code, the boot sector and kernel of an operating system can be grafted together into a *kernel image*, which can be written to the initial sectors of the boot disk, such that the boot sector code is always at the head of the kernel image. Once we have compiled the boot sector described in this section, we can create our kernel image with the following file concatenation command:

```
cat boot.sect.bin kernel.bin > os-image
```

Figure XXX shows a boot sector that will bootstrap our kernel from a disk containing our kernel image, `os-image`.

```
; A boot sector that boots a C kernel in 32-bit protected mode
[org 0x7c00]
KERNEL_OFFSET equ 0x1000 ; This is the memory offset to which we will load our kernel

mov [BOOT_DRIVE], dl ; BIOS stores our boot drive in DL, so it's
                     ; best to remember this for later.

mov bp, 0x9000        ; Set-up the stack.
mov sp, bp

mov bx, MSG_REAL_MODE ; Announce that we are starting
call print_string      ; booting from 16-bit real mode

call load_kernel       ; Load our kernel

call switch_to_pm      ; Switch to protected mode, from which
                     ; we will not return

jmp $

; Include our useful, hard-earned routines
%include "print/print_string.asm"
%include "disk/disk_load.asm"
%include "pm/gdt.asm"
%include "pm/print_string_pm.asm"
%include "pm/switch_to_pm.asm"

[bits 16]

; load_kernel
load_kernel:
    mov bx, MSG_LOAD_KERNEL ; Print a message to say we are loading the kernel
    call print_string

    mov bx, KERNEL_OFFSET   ; Set-up parameters for our disk_load routine, so
    mov dh, 15              ; that we load the first 15 sectors (excluding
    mov dl, [BOOT_DRIVE]    ; the boot sector) from the boot disk (i.e. our
    call disk_load           ; kernel code) to address KERNEL_OFFSET

    ret

[bits 32]
; This is where we arrive after switching to and initialising protected mode.
```

```

BEGIN_PM:

    mov ebx, MSG_PROT_MODE ; Use our 32-bit print routine to
    call print_string_pm   ; announce we are in protected mode

    call KERNEL_OFFSET     ; Now jump to the address of our loaded
                           ; kernel code, assume the brace position,
                           ; and cross your fingers. Here we go!

    jmp $                  ; Hang.

; Global variables
BOOT_DRIVE      db 0
MSG_REAL_MODE   db "Started in 16-bit Real Mode", 0
MSG_PROT_MODE   db "Successfully landed in 32-bit Protected Mode", 0
MSG_LOAD_KERNEL db "Loading kernel into memory.", 0

; Bootsector padding
times 510-($-$$) db 0
dw 0xaa55

```

Before running this command in Bochs, ensure that the Bochs configuration file has the boot disk set to your kernel image file, as in Figure XXX.

```

floppya: 1_44=os-image, status=inserted
boot: a

```

One question you might be wondering is why we loaded as many as 15 segments (i.e.  $512 * 15$  bytes) from the boot disk, when our kernel image was much smaller than this; actually it was less than one sector in size, so to load 1 sector would have done the job. The reason is simply that it does not hurt to read those additional sectors from the disk, even if they have not been initialised with data, but it may hurt when trying to detect that we didn't read enough sectors at this stage when we later add to, and therefore increase the memory footprint size of, our kernel code: the computer would hang without warning, perhaps halfway through a routine that was split across an unloaded sector boundary --- an ugly bug.

Congratulations if an 'X' was displayed in the top-left corner of the screen, since though appearing pointless to the average computer user this signifies a huge step from where we started: we have now boot-strapped into a higher-level language, and can start to worry less about assembly coding and concern ourselves more with how we would like to develop our operating system, and learning a little more about C, of course; but this is the best way to learn C: looking up to it as a higher level language rather than looking down upon it from the perspective of an even higher abstraction, such as Java or a scripting language (e.g. Python, PHP, etc.).

### 5.2.3 Finding Our Way into the Kernel

It was definitely a good idea to start with a very simple kernel, but by doing so we overlooked a potential problem: when we boot the kernel, recklessly we jumped to, and therefore began execution from, the first instruction of the kernel code; but we saw in Section XXX how the C compiler can decide to place code and data wherever it chooses in the output file. Since our simple kernel had a single function, and based on our previous observations of how the compiler generates machine code, we might assume that the first machine code instruction is the first instruction of kernel's entry function, `main`, but suppose our kernel code look like that in Figure XXX.

```
void some_function() {
}

void main() {
    char* video_memory = 0xb8000;
    *video_memory = 'X';
    // Call some function
    some_function();
}
```

Now, the compiler will likely precede the instructions of the intended entry function `main` by those of `some_function`, and since our boot-strapping code will begin execution blindly from the first instruction, it will hit the first `ret` instruction of `some_function` and return to the boot sector code without ever having entered `main`. The problem is, that entering our kernel in the correct place is too dependant upon the ordering of elements (e.g. functions) in our kernel's source code and upon the whims of the compiler and linker, so we need to make this more robust.

A trick that many operating systems use to enter the kernel correctly is to write a very simple assembly routine that is always attached to the start of the kernel machine code, and whose sole purpose is to call the entry function of the kernel. The reason assembly is used is because we know exactly how it will be translated in machine code, and so we can make sure that the first instruction will eventually result in the kernel's entry function being reached.

This is a good example of how the linker works, since we haven't really exploited this important tool yet. The linker takes object files as inputs, then joins them together, but resolves any labels to their correct addresses. For example, if one object file has a piece of code that has a call to a function, `some_function`, defined in another object file, then after the object file's code has been physically linked together into one file, the label `:code:some_function` will be resolved to the offset of wherever that particular routine ended up in the combined code.

Figure XXX shows a simple assembly routine for entering the kernel.

```
; Ensures that we jump straight into the kernel's entry function.
[bits 32]          ; We're in protected mode by now, so use 32-bit instructions.
[extern main]      ; Declare that we will be referencing the external symbol 'main',
                  ; so the linker can substitute the final address
```

```
call main      ; invoke main() in our C kernel
jmp $          ; Hang forever when we return from the kernel
```

You can see from the line `call main` that the code simply calls a function that goes by the name of `main`. But `main` does not exist as a label within this code, since it is expected to exist within one of the other object files, such that it will be resolved to the correct address at link time; this expectance is expressed by the directive `[extern main]`, at the top of the file, and the linker will fail if it doesn't find such a label.

Previously we have compiled assembly into a raw binary format, because we wanted to run it as boot sector code on the CPU, but for this piece of code cannot stand alone, without having that label resolved, so we must compile it as follows as an object file, therefore preserving meta information about the labels it must resolve:

```
$nasm kernel_entry.asm -f elf -o kernel_entry.o
```

The option `-f elf` tells the assembler to output an object file of the particular format Executable and Linking Format (ELF), which is the default format output by out C compiler.

Now, rather than simple linking the `kernel.o` file with itself to create `kernel.bin`, we can link it with `kernel_entry.o`, as follows:

```
$ld -o kernel.bin -Ttext 0x1000 kernel_entry.o kernel.o --oformat binary
```

The linker respects the order of the files we gave to it on the command line, such that the previous command will ensure our `kernel_entry.o` will precede the code in `kernel.o`.

As before, we can reconstruct our kernel image file with the following command:

```
cat boot_sect.bin kernel.bin > os-image
```

Now we can test this in Bochs, but with more reassurance that our boot-block will find its way into the correct entry point of our kernel.

## 5.3 Automating Builds with Make

By now you should be fed up of having to re-type lots of commands, every time you change a piece of code, to get some feedback on a correction or a new idea you tried. Again, programmers have been here before, and have developed a multitude of tools for automating the build process of software. Here we will consider `make`, which is the predecessor of many of these other build tools, and which is used for building, amongst other operating systems and applications, Linux and Minix. The basic principle of `make` is that we specify in a configuration file (usually called `Makefile`) how to convert one file into another, such that the generation of one file may be described to depend on the existence of one or more other file. For example, we could write the following rule in a Makefile, that would tell `make` exactly how to compile a C file into an object file:

```
kernel.o : kernel.c
    gcc -ffreestanding -c kernel.c -o kernel.o
```

The beauty of this is that, in the same directory as the Makefile, we can now type:

```
$make kernel.o
```

which will re-compile our C source file only if `kernel.o` does not exist or has an older file modification time than `kernel.c`. But it is only when we add several inter-dependant rules that we see how `make` can really help us to save time and unnecessary command executions.

```
# Build the kernel binary
kernel.bin: kernel_entry.o kernel.o
    ld -o kernel.bin -Ttext 0x1000 kernel_entry.o kernel.o --oformat binary

# Build the kernel object file
kernel.o : kernel.c
    gcc -ffreestanding -c kernel.c -o kernel.o

# Build the kernel entry object file.
kernel_entry.o : kernel_entry.asm
    nasm kernel_entry.asm -f elf -o kernel_entry.o
```

If we run `make kernel.bin` with the Makefile in Figure XXX, `make` will know that, before it can run the command to generate `kernel.bin`, it must build its two dependencies, `kernel.o` and `kernel_entry.o`, from their source files, `kernel.c` and `kernel_entry.asm`, yeilding the following output of the commands it ran:

```
nasm kernel_entry.asm -f elf -o kernel_entry.o
gcc -ffreestanding -c kernel.c -o kernel.o
ld -o kernel.bin -Ttext 0x1000 kernel_entry.o kernel.o --oformat binary
```

Then, if we run `make` again, we will see that `make` reports that the build target `kernel.bin` is up to date. However, if we modify, say, `kernel.c`, save it, then run `make kernel.bin`, we will see that only the necessary commands are run by `make`, as follows:

```
gcc -ffreestanding -c kernel.c -o kernel.o
ld -o kernel.bin -Ttext 0x1000 kernel_entry.o kernel.o --oformat binary
```

To reduce repetition in, and therefore improve ease of maintenance of, our Makefile, we can use the special makefile variables `$<`, `$@`, and `$^` as in Figure XXX.

```
# $^ is substituted with all of the target's dependency files
kernel.bin: kernel_entry.o kernel.o
    ld -o kernel.bin -Ttext 0x1000 $^ --oformat binary

# $< is the first dependency and $@ is the target file
kernel.o : kernel.c
    gcc -ffreestanding -c $< -o $@

# Same as the above rule.
kernel_entry.o : kernel_entry.asm
    nasm $< -f elf -o $@
```

It is often useful to specify targets that are not actually real targets, in that they do not generate files. A common use of such phoney targets is to make a `clean` target, so that when we run `make clean`, all of the generated files are deleted from the directory, leaving only the source files, as in Figure XXX.

```
clean:
    rm *.bin *.o
```

Cleaning your directory in this way is useful if you'd like to distribute only the source files to a friend, put the directory under version control, or if you'd like to test that modifications of your makefile will correctly build all targets from scratch.

If `make` is run without a target, the first target in the main file is taken to be the default, so you often see a phoney target such as `all` at the top of Makefile as in Figure XXX.

```
# Default make target.
all: kernel.bin

# $^ is substituted with all of the target's dependency files
kernel.bin: kernel_entry.o kernel.o
    ld -o kernel.bin -Ttext 0x1000 $^ --oformat binary

# $< is the first dependency and $@ is the target file
kernel.o : kernel.c
    gcc -ffreestanding -c $< -o $@

# Same as the above rule.
kernel_entry.o : kernel_entry.asm
    nasm $< -f elf -o $@
```

Note that, by giving `kernel.bin` as a dependency to the `all` target, we ensure that `kernel.bin` and all of its dependencies are built for the default target.

We can now put all of the commands for building our kernel and the loadable kernel image into a useful makefile (see Figure XXX), that will allow us to test changes or corrections to our code in Bochs simply by typing `make run`.

```
all: os-image

# Run bochs to simulate booting of our code.
run: all
    bochs

# This is the actual disk image that the computer loads,
# which is the combination of our compiled bootsector and kernel
os-image: boot_sect.bin kernel.bin
    cat $^ > os-image
```

```

# This builds the binary of our kernel from two object files:
# - the kernel_entry, which jumps to main() in our kernel
# - the compiled C kernel
kernel.bin: kernel_entry.o kernel.o
    ld -o kernel.bin -Ttext 0x1000 $^ --oformat binary

# Build our kernel object file.
kernel.o : kernel.c
    gcc -ffreestanding -c $< -o $@

# Build our kernel entry object file.
kernel_entry.o : kernel_entry.asm
    nasm $< -f elf -o $@

# Assemble the boot sector to raw machine code
# The -I options tells nasm where to find our useful assembly
# routines that we include in boot_sect.asm
boot_sect.bin : boot_sect.asm
    nasm $< -f bin -I '../16bit/' -o $@

# Clear away all generated files.
clean:
    rm -fr *.bin *.dis *.o os-image *.map

# Disassemble our kernel - might be useful for debugging.
kernel.dis : kernel.bin
    ndisasm -b 32 $< > $@

```

### 5.3.1 Organising Our Operating System's Code Base

We have now arrived at a very simple C kernel, that prints out an 'X' in the corner of the screen. The very fact that the kernel was compiled into 32-bit instructions and has successfully been executed by the CPU means that we have come far; but it is now time to prepare ourselves for the work ahead. We need to establish a suitable structure for our code, accompanied by a makefile that will allow us easily to add new source files with new features to our operating system, and to check those additions incrementally with an emulator such as Bochs.

Similarly to kernels such as Linux and Minix, we can organise our code base into the following folders:

- **boot**: anything related to booting and the boot sector can go in here, such as `boot_sect.asm` and our boot sector assembly routines (e.g. `print_string.asm`, `gdt.asm`, `switch_to_pm.asm`, etc.).
- **kernel**: the kernel's main file, `kernel.c`, and other kernel related code that is not device driver specific will go in here.
- **drivers**: any hardware specific driver code will go in here.

Now, within our makefile, rather than having to specify every single object file that we would like to build (e.g. `kernel/kernel.o`, `drivers/screen.o`, `drivers/keyboard.o`, etc.), we can use a special *wildcard* declaration as follows:

```
# Automatically expand to a list of existing files that
# match the patterns
C_SOURCES = $(wildcard kernel/*.c drivers/*.c)
```

Then we can convert the source filenames into object filenames using another `make` declaration, as follows:

```
# Create a list of object files to build, simple by replacing
# the '.c' extension of filenames in C_SOURCES with '.o'
OBJ = ${C_SOURCES:.c=.o}
```

Now we can link the kernel object files together, to build the kernel binary, as follows:

```
# Link kernel object files into one binary, making sure the
# entry code is right at the start of the binary.
kernel.bin: kernel/kernel_entry.o ${OBJ}
    ld -o $@ -Ttext 0x1000 $^ --oformat binary
```

A feature of `make` that will go hand-in-hand with our dynamic inclusion of object files is *pattern rules*, which tell `make` how to build one file type from another based on simple pattern machine of the filename, as follows:

```
# Generic rule for building 'somefile.o' from 'somefile.c'
%.o : %.c
    gcc -ffreestanding -c $< -o $@
```

The alternative to this would be much repetition, as follows:

```
kernel/kernel.o : kernel/kernel.c
    gcc -ffreestanding -c $< -o $@

drivers/screen.o : drivers/screen.c
    gcc -ffreestanding -c $< -o $@

drivers/keyboard.o : drivers/keyboard.c
    gcc -ffreestanding -c $< -o $@

...
```

Great, now that we understand `make` sufficiently, we can progress to develop our kernel, without having to re-type lots of commands, over and over, to check if something is working correctly. Figure XXX shows a complete makefile that will be suitable for progressing with our kernel.

```
# Automatically generate lists of sources using wildcards.
C_SOURCES = $(wildcard kernel/*.c drivers/*.c)
HEADERS = $(wildcard kernel/*.h drivers/*.h)

# TODO: Make sources dep on all header files.

# Convert the *.c filenames to *.o to give a list of object files to build
OBJ = ${C_SOURCES:.c=.o}

# Default build target
all: os-image
```

```

# Run bochs to simulate booting of our code.
run: all
    bochs

# This is the actual disk image that the computer loads
# which is the combination of our compiled bootsector and kernel
os-image: boot/boot_sect.bin kernel.bin
    cat $^ > os-image

# This builds the binary of our kernel from two object files:
# - the kernel_entry, which jumps to main() in our kernel
# - the compiled C kernel
kernel.bin: kernel/kernel_entry.o ${OBJ}
    ld -o $@ -Ttext 0x1000 $^ --oformat binary

# Generic rule for compiling C code to an object file
# For simplicity, we C files depend on all header files.
%.o : %.c ${HEADERS}
    gcc -ffreestanding -c $< -o $@

# Assemble the kernel_entry.
%.o : %.asm
    nasm $< -f elf -o $@

%.bin : %.asm
    nasm $< -f bin -I '../.. /16bit/' -o $@

clean:
    rm -fr *.bin *.dis *.o os-image
    rm -fr kernel/*.o boot/*.bin drivers/*.o

```

## 5.4 C Primer

C has a few quirks that can unsettle a new programmer of the language.

### 5.4.1 The Pre-processor and Directives

Before a C file is compiled into an object file, a pre-processor scans it for pre-processor directives and variables, and then usually substitutes them with code, such as macros and values of constants, or with nothing at all. The pre-processor is not essential for compiling C code, but serves rather to offer some convenience that makes the code more manageable.

```

#define PI 3.141592
...
float radius = 3.0;
float circumference = 2 * radius * PI;
...

```

The pre-processor would output the following code, ready for compilation:

```
...
float radius = 3.0;
float circumference = 2 * radius * 3.141592;
...
```

The pre-processor is also useful for outputting conditional code, but not conditional in the sense that a decision is made at run-time, like with an `if` statement, rather in the sense of compile-time. For example, consider the following use of pre-processor directives for the inclusion or exclusion of debugging code:

```
...
#ifdef DEBUG
print("Some debug message\n");
#endif
...
```

Now, if the pre-processor variable `DEBUG` has been defined, such debugging code will be included; otherwise, not. A variable may be defined on the command line when compiling the C file as follows:

```
$gcc -DDEBUG -c some_file.c -o some_file.o
```

Such command line variable declarations are often used for compile-time configuration of applications, and especially operating systems, which may include or exclude whole sections of code, perhaps to reduce the memory footprint of the kernel on a small embedded device.

## 5.4.2 Function Declarations and Header Files

When the compiler encounters a call to a function, that may or may not be defined in the file being compiled, it may make incorrect assumptions and produce incorrect machine code instructions if it has not yet encountered a description of the functions return type and arguments. Recall from Section XXX that the compiler must prepare the stack for variables that it passes to a function, but if the stack is not what the function expects, then the stack may become corrupted. For this reason, it is important that at least a declaration of the function's interface, if not the entire function definition, is given before it is used. This declaration is known as a function's *prototype*.

```
int add(int a, int b) {
    return a + b;
}

void main() {
    // This is okay, because our compiler has seen the full
    // definition of add.
    int result = add(5, 3);

    // This is not okay, since compiler does not know the return
    // type or anything about the arguments.
    result = divide(34.3, 12.76);

    // This is not okay, because our compiler knows nothing about
```

```
// this function's interface.
int output = external_function(5, "Hello", 4.5);
}

float divide(float a, float b) {
    return a / b;
}
```

This can be fixed as follows:

```
// These function prototypes inform the compiler about
// the function interfaces.
float divide(float a, float b); // <-- note the semi-colon
int external_function(int a, char* message, float b);

int add(int a, int b) {
    return a + b;
}

void main() {
    // This is okay, because our compiler has seen the full
    // definition of add.
    int result = add(5, 3);

    // This is okay now: compiler knows the interface.
    result = divide(34.3, 12.76);

    // This is okay now: compiler knows the interface.
    int output = external_function(5, "Hello", 4.5);
}

float divide(float a, float b) {
    return a / b;
}
```

Now, since some functions will be called from code compiled into other object files, they will also need to declare identical prototypes of those functions, which would lead to a lot of duplicated prototype declarations, which is difficult to maintain. For this reason, many C programs use the `#include` pre-processor directive to insert common code that contains the required prototypes prior to compilation. This common code is known as a *header file*, which we can think of as the interface to the compiled object file, and which is used as follows.

Sometimes, one header file may include another, so it is important not to re-define the same code...

- Casting types

## Chapter 6

---

# Developing Essential Device Drivers and a Filesystem

---

INTRO PART.

## 6.1 Hardware Input/Output

By writing to the screen we have actually already encountered a friendlier form of hardware I/O, known as memory-mapped I/O, whereby data written directly to a certain address range in main memory is written to the device's internal memory buffer, but now it is time to understand more about this interaction between CPU and hardware.

Let's take the now-popular TFT monitor as an example. The screen's surface is divided up into a matrix of backlit cells. By containing a layer of liquid crystals sandwiched between polarised film, the amount of light passing through each cell can be varied by the application of an electric field, since liquid crystals have the property that, when subjected to an electrical field, their orientation may be altered in a consistent manner; as the orientation of the crystals changes, they alter the light wave's direction of vibration, such that some of the light will be blocked by the polarised film at the screen's surface. For a colour display, each cell is further divided into three areas that are overlaid with filters for red, blue, and green [?].

So it is the hardware's job to ensure that the appropriate cells, or sub-cell colour areas, get subjected to appropriate electrical field to reconstruct the desired image on the screen. This side of hardware is best left to specialist electronic engineers, but there will be a *controller* chip, ideally with well defined functionality that is described in the chip's datasheet, on the device or motherboard with which the CPU can interact to direct the hardware. In reality, for reasons of backward compatibility, TFT monitors usually emulate older CRT monitors, and so can be driven by the motherboard's standard VGA controller, which generates a complex analog signal that directs an electron beam to scan across the phosphor-coated screen, and since there isn't really a CRT beam to direct, the TFT monitor cleverly interprets this signal as a digital image.

Internally, controller chips usually have several registers that can be read, written or both by the CPU, and it is the state of these registers that tell the controller what to do (e.g. what pins to set high or low to drive the hardware, or what internal function to perform). As an example, from the datasheet of Intel's widely used [82077AA](#) single-chip floppy disk controller [?], we see there is a pin (pin 57, labelled [MEO](#)) that drives the motor of the first floppy disk device (since a single controller can drive several such devices): when the pin is on, the motor spins; when off, the motor does not spin. The state of this particular pin is directly linked to a particular bit of the controller's internal register named the Digital Output Register ([DOR](#)). The state of that register can then be set by setting a value, with the appropriate bit set (bit 4, in this case), across the chip's data pins, labelled [DB0--DB7](#), and using the chip's register selection pins, [A0--A2](#), to select the [DOR](#) register by its internal address [0x2](#).

### 6.1.1 I/O Buses

Although historically the CPU would talk directly to device controllers, with ever increasing CPU speeds, that would require the CPU artificially to slow down to the same speed as the slowest device, so it is more practical for the CPU to issue I/O instructions directly to the controller chip of a high-speed, top-level *bus*. The bus controller is then responsible for relaying, at a compatible rate, the instructions to a particular device's controller. Then to avoid the top-level bus having to slow down for slower devices, the controller of another bus technology may be added as a device, such that we arrive at the hierarchy of buses found in modern computers [?].

### 6.1.2 I/O Programming

So the question is, how do we read and write the registers of our device controllers (i.e. tell our devices what to do) programatically? In Intel architecture systems the registers of device controllers are mapped into an I/O address space, that is separate from the main memory address space, then variants of the I/O instructions [in](#) and [out](#) are used to read and write data to the I/O addresses that are mapped to specific controller registers. For example, the floppy disk controller described earlier usually has its [DOR](#) register mapped to I/O address [0x3F2](#), so we could switch on the motor of the first drive with the following instructions:

```
mov dx, 0x3f2      ; Must use DX to store port address
in al, dx          ; Read contents of port (i.e. DOR) to AL
or al, 00001000b   ; Switch on the motor bit
out dx, al         ; Update DOR of the device.
```

In older systems, such as the Industry Standard Architecture (ISA) bus, the port addresses would be statically assigned to the devices, but with modern plug-and-play buses, such as Peripheral Component Interconnect (PCI), BIOS can dynamically allocate I/O address to most devices before booting the operating system. Such dynamic allocation requires devices to communicate configuration information over the bus to describe the hardware such as: how many I/O ports are required to be reserved for the registers; how much memory-mapped space is required; and a unique ID of the hardware type, to allow appropriate drivers to be found later by the operating system [?].

A problem with port I/O is that we cannot express these low-level instructions in C language, so we have to learn a little bit about *inline* assembly: most compilers allow you to inject snippets of assembly code into the body of a function, with gcc implementing this as follows:

```
unsigned char port_byte_in(unsigned short port) {
    // A handy C wrapper function that reads a byte from the specified port
    // "=a" (result) means: put AL register in variable RESULT when finished
    // "d" (port) means: load EDX with port
    unsigned char result;
    __asm__("in %%dx, %%al" : "=a" (result) : "d" (port));
    return result;
}
```

Note that the actual assembly instruction, `in %%dx, %%al`, looks a little strange to us, since gcc adopts a different assembly syntax (known as GAS), where the target and destination operands are reversed with respect to the syntax of our more familiar nasm syntax; also, `%` is used to denote registers, and this requires an ugly `%%`, since `%` is an escape character of the C compiler, and so `%%` means: escape the escape character, so that it will appear literally in the string.

Since these low-level port I/O functions will be used by most hardware drivers in our kernel, let's collect them together into the file `kernel/low_level.c`, which we can define as in Figure XXX.

```
unsigned char port_byte_in(unsigned short port) {
    // A handy C wrapper function that reads a byte from the specified port
    // "=a" (result) means: put AL register in variable RESULT when finished
    // "d" (port) means: load EDX with port
    unsigned char result;
    __asm__("in %%dx, %%al" : "=a" (result) : "d" (port));
    return result;
}

void port_byte_out(unsigned short port, unsigned char data) {
    // "a" (data) means: load EAX with data
    // "d" (port) means: load EDX with port
    __asm__("out %%al, %%dx" : : "a" (data), "d" (port));
}

unsigned short port_word_in(unsigned short port) {
    unsigned short result;
    __asm__("in %%dx, %%ax" : "=a" (result) : "d" (port));
    return result;
}

void port_word_out(unsigned short port, unsigned short data) {
    __asm__("out %%ax, %%dx" : : "a" (data), "d" (port));
}
```

### 6.1.3 Direct Memory Access

Since port I/O involves reading or writing individual bytes or words, the transfer of large amounts of data between a disk device and memory could potentially take up a great deal of better-spent CPU time. This issue has necessitated a means for the CPU to pass over this tedious task to someone else, a direct memory access (DMA) controller [?].

A good analogy of DMA is that of an architect wanting to move a wall from one place to another. The architect knows exactly what is to be done but has other important things to consider other than shifting each brick, and so instructs a builder to move the bricks, one by one, and to alert (i.e. interrupt) him when either the wall is finished or if there was some error that is stopping the wall from being finished.

## 6.2 Screen Driver

So far, our kernel is capable of printing an 'X' in the corner of the screen, which, whilst is sufficient to let us know our kernel has been successfully loaded and executed, doesn't tell us much about what is happening on the computer.

We know that we can have characters displayed on the screen by writing them somewhere within the display buffer at address `0xb8000`, but we don't want to keep having to worry about that sort of low-level manipulation throughout our kernel. It would be much nicer if we could create an abstraction of the screen that would allow us to write `print('Hello')`, and perhaps `clear_screen()`; and if it could scroll when we printed beyond the last display line, that would be icing on the cake. Not only would this sort of abstraction make it easier to display information within other code of our kernel, but it would allow us to easily substitute one display driver for another at a later date, perhaps if a certain computer could not support the colour VGA text mode that we currently assume.

### 6.2.1 Understanding the Display Device

Compared with some of the other hardware that we will soon look at, the display device is fairly straightforward, since, as a memory-mapped device, we can get by without understanding anything about control messages and hardware I/O. However, a useful device of the screen that requires I/O control (i.e. via I/O ports) to manipulate is the *cursor*, that flashes to mark the next position that a character will be written to on the screen. This is useful for a user, since it can draw their attention to a prompt to enter some text, but we will also use it as an internal marker, whether the cursor is visible or not, so that a programmer does not always have to specify the coordinates of where on the screen a string is to be displayed, for example: if we write `print('hello')`, each character will be written to.

### 6.2.2 Basic Screen Driver Implementation

Although we could write all of this code in `kernel.c`, that contains the kernel's entry function, `main()`, it is good to organise such functionality-specific code into it's own file, which can be compiled and linked to our kernel code, ultimately with the same effect as putting it all into one file. Let's create a new driver implementation file, `screen.c`, and

a driver interface file, `screen.h`, in our `drivers` folder. Due to our use of *wildcard* file inclusion in our makefile, `screen.c` will (as will any other C files placed in that folder) be automatically compiled and linked to our kernel.

Firstly, let's define the following constants in `screen.h`, to make our code more readable:

```
#define VIDEO_ADDRESS 0xb8000
#define MAX_ROWS 25
#define MAX_COLS 80
// Attribute byte for our default colour scheme.
#define WHITE_ON_BLACK 0x0f

// Screen device I/O ports
#define REG_SCREEN_CTRL 0x3D4
#define REG_SCREEN_DATA 0x3D5
```

Then, let's consider how we would write a function, `print_char(...)`, that displays a single character at a specific column and row of the screen. We will use this function internally (i.e. privately), within our driver, such that our driver's public interface functions (i.e. the functions that we would like external code to use) will build upon it. We now know that video memory is simply a specific range of memory addresses, where each character cell is represented by two bytes, the first byte is the ASCII code of the character, and the second byte is an attribute byte, that allows us to set a colourscheme of the character cell. Figure XXX shows how we could define such a function, by making use of some other functions that we will define: `get_cursor()`, `set_cursor()`, `get_screen_offset()`, and `handle_scrolling()`.

```
/* Print a char on the screen at col, row, or at cursor position */
void print_char(char character, int col, int row, char attribute_byte) {
    /* Create a byte (char) pointer to the start of video memory */
    unsigned char *vidmem = (unsigned char *) VIDEO_ADDRESS;

    /* If attribute byte is zero, assume the default style. */
    if (!attribute_byte) {
        attribute_byte = WHITE_ON_BLACK;
    }

    /* Get the video memory offset for the screen location */
    int offset;
    /* If col and row are non-negative, use them for offset. */
    if (col >= 0 && row >= 0) {
        offset = get_screen_offset(col, row);
    } /* Otherwise, use the current cursor position. */
    else {
        offset = get_cursor();
    }

    // If we see a newline character, set offset to the end of
    // current row, so it will be advanced to the first col
    // of the next row.
    if (character == '\n') {
```

```

    int rows = offset / (2*MAX_COLS);
    offset = get_screen_offset(79, rows);
    // Otherwise, write the character and its attribute byte to
    // video memory at our calculated offset.
} else {
    vidmem[offset] = character;
    vidmem[offset+1] = attribute_byte;
}

// Update the offset to the next character cell, which is
// two bytes ahead of the current cell.
offset += 2;
// Make scrolling adjustment, for when we reach the bottom
// of the screen.
offset = handle_scrolling(offset);
// Update the cursor position on the screen device.
set_cursor(offset);
}

```

Let's tackle the easiest of these functions first: `get_screen_offset`. This function will map row and column coordinates to the memory offset of a particular display character cell from the start of video memory. The mapping is straightforward, but we must remember that each cell holds two bytes. For example, if I want to set a character at row 3, column 4 of the display, then the character cell of that will be at a (decimal) offset of 488 ( $(3 * 80 \text{ (i.e. the row width)} + 4) * 2 = 488$ ) from the start of video memory. So our `get_screen_offset` function will look something like that in Figure XXX.

```

// This is similar to get_cursor, only now we write
// bytes to those internal device registers.
port_byte_out(REG_SCREEN_CTRL, 14);
port_byte_out(REG_SCREEN_DATA, (unsigned char)(offset >> 8));
port_byte_out(REG_SCREEN_CTRL, 15);

```

Now let's look at the cursor control functions, `get_cursor()` and `set_cursor()`, which will manipulate the display controller's registers via a set of I/O ports. Using the specific video devices I/O ports to read and write its internal cursor-related registers, the implementation of these functions will look something like that in Figure XXX.

```

    cursor_offset -= 2*MAX_COLS;

    // Return the updated cursor position.
    return cursor_offset;
}

int get_cursor() {
    // The device uses its control register as an index
    // to select its internal registers, of which we are
    // interested in:

```

```
// reg 14: which is the high byte of the cursor's offset
// reg 15: which is the low byte of the cursor's offset
// Once the internal register has been selected, we may read or
// write a byte on the data register.
port_byte_out(REG_SCREEN_CTRL, 14);
int offset = port_byte_in(REG_SCREEN_DATA) << 8;
port_byte_out(REG_SCREEN_CTRL, 15);
offset += port_byte_in(REG_SCREEN_DATA);
// Since the cursor offset reported by the VGA hardware is the
// number of characters, we multiply by two to convert it to
// a character cell offset.
return offset*2;
}

void set_cursor(int offset) {
    offset /= 2; // Convert from cell offset to char offset.
    // This is similar to get_cursor, only now we write
    // bytes to those internal device registers.
```

So now we have a function that will allow us to print a character at a specific location of the screen, and that function encapsulates all of the messy hardware specific stuff. Usually, we will not want to print each character to the screen, but rather a whole string of characters, so let's create a friendlier function, `print_at(...)`, that takes a pointer to the first character of a string (i.e. a `char *`) and prints each subsequent character, one after the other, from the given coordinates. If the coordinates `(-1,-1)` are passed to the function, then it will start printing from the current cursor location. Our `print_at(...)` function will look something like that in Figure XXX.

```
void print_at(char* message, int col, int row) {
    // Update the cursor if col and row not negative.
    if (col >= 0 && row >= 0) {
        set_cursor(get_screen_offset(col, row));
    }
    // Loop through each char of the message and print it.
    int i = 0;
    while(message[i] != 0) {
        print_char(message[i++], col, row, WHITE_ON_BLACK);
    }
}
```

And purely for convenience, to save us from having to type `print_at('hello', -1,-1)`, we can define a function, `print`, that takes only one argument as in Figure XXX.

```
void print(char* message) {
    print_at(message, -1, -1);
}
```

Another useful, but not too difficult function, is `clear_screen(...)`, which will allow us to tidy up our screen by writing blank characters at every position. Figure XXX shows how we might implement such a function.

```
void clear_screen() {
    int row = 0;
    int col = 0;

    /* Loop through video memory and write blank characters. */
    for (row=0; row<MAX_ROWS; row++) {
        for (col=0; col<MAX_COLS; col++) {
            print_char(' ', col, row, WHITE_ON_BLACK);
        }
    }

    // Move the cursor back to the top left.
    set_cursor(get_screen_offset(0, 0));
}
```

### 6.2.3 Scrolling the Screen

If you expected the screen to scroll automatically when your cursor reached the bottom of the screen, then your brain must have lapsed back into higher-level computer land. This can be forgiven, because screen scrolling seems like such a natural thing that we simply take for granted; but working at this level, we have complete control over the hardware, and so must implement this feature ourselves.

In order to make the screen appear to scroll when we reach the bottom, we must move each character cell upwards by one row, and then clear the last row, ready for writing the new row (i.e. the row that would otherwise have been written beyond the end of the screen). This means the the top row will be overwritten by the second row, and so the top row will be lost forever, which we will not concern ourselves with, since our aim is to allow the user to see the most recent log of activity on their computer.

A nice way to implement scrolling is to call a function, which we will define as `handle_scrolling`, immediately after incrementing the cursors position in our `print_char`. The the roll of `handle_scrolling`, then, is to ensure that, whenever the cursor's video memory offset is incremented beyond the last row of the screen, the rows are scrolled and the cursor is repositioned within the last visible row (i.e. the new row).

Shifting a row equates to copying all of its bytes --- two bytes for each of the 80 character cells in a row --- to the address of the previous row. This is a perfect opportunity for adding a general purpose `memory_copy` function to our operating system. Since we are likely to use such a function in other areas of our OS, let's add it to the file `kernel/util.c`. Our `memory_copy` function will take addresses of the source and destination and the number of bytes to copy, then, with a loop, will copy each byte as in Figure XXX.

```
/* Copy bytes from one place to another. */
void memory_copy(char* source, char* dest, int no_bytes) {
```

```
int i;
for (i=0; i<no_bytes; i++) {
    *(dest + i) = *(source + i);
}
}
```

Now we can use `memory_copy`, as in Figure XXX, to scroll our screen.

```
/* Advance the text cursor, scrolling the video buffer if necessary. */
int handle_scrolling(int cursor_offset) {

    // If the cursor is within the screen, return it unmodified.
    if (cursor_offset < MAX_ROWS*MAX_COLS*2) {
        return cursor_offset;
    }

    /* Shuffle the rows back one. */
    int i;
    for (i=1; i<MAX_ROWS; i++) {
        memory_copy(get_screen_offset(0,i) + VIDEO_ADDRESS,
                    get_screen_offset(0,i-1) + VIDEO_ADDRESS,
                    MAX_COLS*2);
    }

    /* Blank the last line by setting all bytes to 0 */
    char* last_line = get_screen_offset(0,MAX_ROWS-1) + VIDEO_ADDRESS;
    for (i=0; i < MAX_COLS*2; i++) {
        last_line[i] = 0;
    }

    // Move the offset back one row, such that it is now on the last
    // row, rather than off the edge of the screen.
    cursor_offset -= 2*MAX_COLS;

    // Return the updated cursor position.
    return cursor_offset;
}
```

## 6.3 Handling Interrupts

## 6.4 Keyboard Driver

## 6.5 Hard-disk Driver

## 6.6 File System

## *Chapter 7*

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# **Implementing Processes**

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**7.1 Single Processing**

**7.2 Multi-processing**

## *Chapter 8*

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# Summary

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## Bibliography

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